

LESSON 1: HUMAN-COMPUTER INTERACTION

Human-Computer Interaction (HCI) is a multidisciplinary field that focuses on the study of the design of computers and, in particular, the interaction that occurs between humans and computers.

Initially, HCI focused only on computers, but this has changed over the years to currently include almost all forms of information technologies. Understanding how humans interact with computers and other technologies has greatly changed our lives. It has made computers and technology more user-friendly and intuitive, which has greatly broadened the range of people who can comfortably use these technologies.

HUMAN-COMPUTER INTERFACE

A system's interface is the boundary across which it interacts with the outside world. The system receives input and delivers output across its interface, which is physically determined by the system's input and output devices.

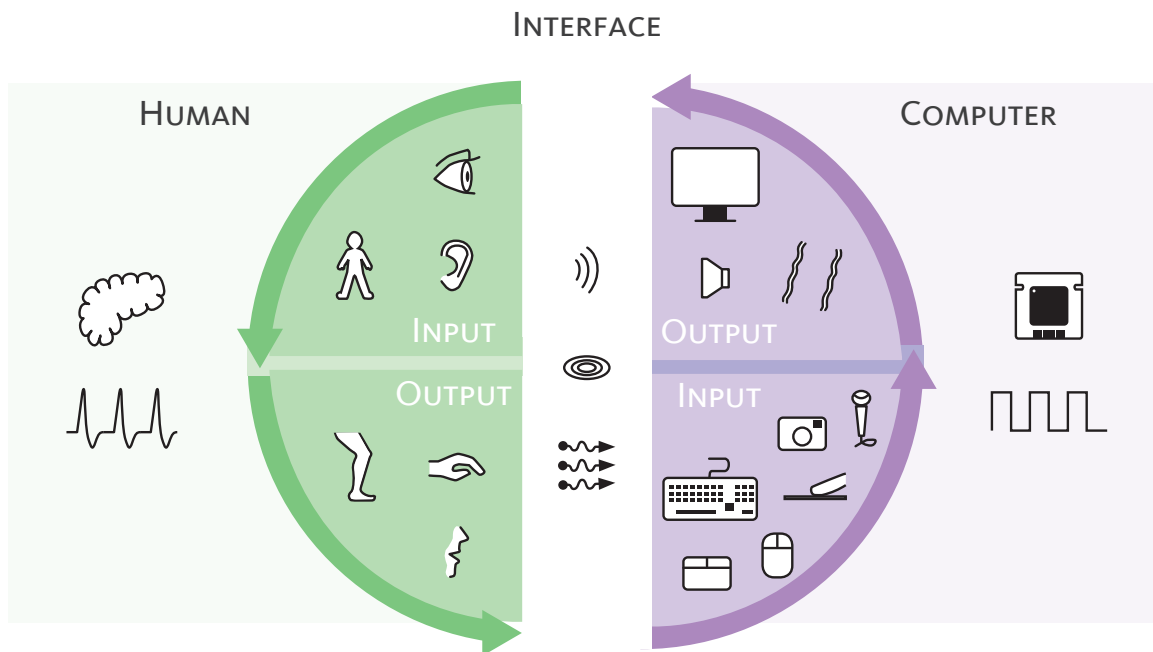
Two systems can interact if their interfaces are compatible and allow them to exchange information. Humans have designed computer interfaces to be compatible with their own. This allows humans to exchange information with computers and to interact with them. So when a human uses a computer, the boundary where the two systems meet and exchange signals is referred to as the human-computer interface.

THE HUMAN-COMPUTER INTERFACE IS THE BOUNDARY ACROSS
WHICH THE TWO SYSTEMS INTERACT.

INPUT/OUTPUT LOOP

The input/output (I/O) loop is fundamental to human-computer interaction. The interface between the two systems allows signals to be passed back and forth. The output of one system becomes the input of the other, which in turn generates an output. This process repeats itself, creating an I/O loop across the interface. The I/O loop perpetuates until one of two conditions arises: 1) the task is completed, or 2) the loop is interrupted for some reason.

HUMAN-COMPUTER INTERACTION IS A LOOP OF INPUTS AND OUTPUTS BEING EXCHANGED BY THE TWO SYSTEMS.



The human-computer interface is established by the input and output components of each system. The user receives a signal emitted from the computer. The sensory organs convert it to action potentials that the human can process and then accordingly respond. The human's response, or output, then becomes the computer's input. The incoming signal emitted by the human is converted to electronic digital signals, which the computer uses to process and respond, creating a loop of inputs and outputs.

EXAMPLE 3.1

When a user enters text into a word processor, the keyboard sends the input to the computer, which processes the input and displays the output on the screen. The user can then see the output and make changes, which are then sent back to the computer. The process repeats in a loop until the desired result is achieved, which is completing the essay, or the loop is interrupted by the user being called for dinner.

PERIPHERALS

Peripheral devices are input and output devices that can be connected to a computer, extending the human-computer interface. They augment an existing functionality or add a new one that the computer could not previously perform. Peripheral devices can act as input, output or as both input and output devices (I/O devices).

**PERIPHERAL DEVICES EXPAND THE HUMAN-COMPUTER INTERFACE
AND PROVIDE ADDITIONAL FUNCTIONALITIES.**

EXAMPLE 3.2

The audio headset is a peripheral output device. It connects to a computer and provides the user with privacy and clarity of sound. In some cases, the headset has a built-in microphone, making it an I/O device that can provide both input and output of sound. This device enhances an existing functionality by bringing the input and output devices closer to the mouth and ears of the user. This new interface provides a more direct and clear transmission of sound signals.

EXAMPLE 3.3

Another example of a peripheral device is the projector, a peripheral output device commonly used in classrooms. This technology outputs light to a much wider area than the computer screen, allowing for a much larger audience of viewers, making the interface not just with one person but several. The smartboard is another classroom peripheral device. It can both display content and allow users to manipulate it directly; it is an I/O device.

There are several peripheral devices that connect with a computer and extend its interface. Some devices connect through wired connections, while others have wireless connections.

Input Devices	Camera - Webcam - Barcode reader - Scanner - Game controller - Microphone
Output Devices	Printer - 3D Printer - Plotter - Projector - Earphones - Speakers - Television
I/O Devices	Headset - Smartboard

By extending the human-computer interface through peripheral devices, users can experience more interactions with computers, allowing them to perform a wide range of tasks across various contexts, such as work, education, entertainment and communication.

It is important to note that each of these devices has its own software, programs that contain the instructions that allow the peripheral device to exchange information with the computer. While this software may not be directly visible in the user interface, it does perform critical work behind the scenes. Without the software, the hardware will not work.

SMART MACHINES

Embedded computers have impacted our world significantly, with processors and other digital components embedded in almost every machine, ranging from home appliances to automobiles.

Embedded computers allow us to enhance the machine's interface as well as provide additional functions that can be performed using the machine. Therefore, smart machines have an interface that provides an enhanced level of functionality and usability, ensuring that the machine performs its intended tasks to the satisfaction of its user.

THE HUMAN-MACHINE INTERFACE IS THE INTERFACE ACROSS
WHICH HUMANS INTERACT WITH MACHINES.

THE SMARTPHONE

The first cellphone was released in 1973 by Motorola, the DynaTAC 8000X. It was large, weighing about 1 kg and costing over \$3,900. As a device, it was purely a phone and could only make and receive calls.

This changed in 1996 with the release of the Nokia 9000 Communicator, the first phone to have an embedded computer. This cellphone provided internet capability, with web browsing and emailing as functionalities. However, it was not until 2007 that the Apple iPhone became the first smartphone to achieve success in the market.

Nowadays, smartphones are no longer just cellphones used for the sole purpose of making calls. Instead, smartphones have become general-purpose machines capable of performing a wide range of tasks. It is now considered a personal computer and not just a smartphone.

THE CASH REGISTER

A cash register is a common example of a machine that contains an embedded computer. The human-machine interface of the register comprises:

Input Devices

Barcode scanner	It is used to scan the barcode on items. The barcode contains information like the name of the product and its price, which is input into the system.
Keyboard	It is used to input information, such as the price of items and method of payment.

Output Devices

Drawer	It is used to store cash. An actuator moves and releases the drawer, allowing it to spring forward and open.
Printer	It is used to print receipts for customers. There are two actuators for this output device: 1) a motor to turn the role of paper and 2) a printhead that moves across printing line by line.
Screen	It is used to show the transaction information, such as the price of items, total cost and change due.

Unlike personal computers that are intended to have a general purpose, the cash register is designed as a machine that serves a specific purpose. Therefore, its interface must be optimized to allow humans to perform the required tasks that serve that purpose. As such, the extent of the interaction between human and machine is limited to the tasks that need to occur at a point of sale, such as identifying items and prices, receiving payments and printing receipts.

ROBOTS

A robot is a machine capable of performing a variety of tasks autonomously or under varying degrees of control by an operator. A robot must interact with the world and humans; as such, it typically consists of input devices in the form of sensors, output devices in the form of actuators that provide movement across joints and a body that holds everything together. The design of the robot and its interface is dictated by the purpose it must serve.

ROBOTS ARE MACHINES CAPABLE OF PERFORMING TASKS AUTONOMOUSLY OR WITH SOME HUMAN INTERVENTION.

Robots can be categorized depending on their degree of autonomy, mobility and purpose.

- **Autonomy:** refers to the degree to which a robot can act independently. An autonomous robot can make decisions on its own without human intervention. A semi-autonomous robot can make most decisions on its own. In comparison, a non-autonomous robot relies on a human operator to control its actions. Therefore, the embedded computer in each of these three cases has varying degrees of decision-making capabilities that can vary from making all decisions on its own to simply assisting in coordinating input and output, leaving the decisions to the human operator.
- **Mobility:** refers to a robot's capability of moving and changing location. Some robots are stationary, such as the robotic arms used in car factories, while other robots are mobile. Mobile robots have varying degrees of mobility. Some can only negotiate a smooth, flat surface, such as robotic vacuum cleaners. In contrast others, such as some advanced humanoid robots, can walk outdoors on uneven terrain, run, jump and even do somersaults.
- **Purpose:** refers to the task or set of tasks a robot is intended to perform. Some tasks are menial and highly repetitive, such as those required in

manufacturing. While other tasks require entering difficult or dangerous locations, such as exploring underwater caves or finding people trapped by fire. Still, other robots, humanoid robots, are designed to serve the purpose of satisfying the engineering achievement of creating machines very much in our own likeness.

Robots have brought about both benefits and concerns. On the one hand, robots have taken over jobs that are unpleasant or even unsafe for humans. Menial and repetitive tasks tend to make jobs unpleasant, and robots have been designed to perform such tasks as:

1. Cleaning floors and windows.
2. Sorting and organizing items in warehouses.
3. Tightening screws or fastening components on assembly lines.
4. Packaging products in factories.

Robots have also taken over jobs and tasks that are dangerous, thereby keeping humans out of harm's way.

1. Handling hazardous materials or chemicals in environments such as nuclear power plants or chemical facilities.
2. Conducting underwater inspections or repairs in deep-sea environments.
3. Performing tasks in areas with high levels of radiation or toxic substances.
4. Exploring hostile or extreme environments like volcanoes, disaster sites and even other planets.
5. Conducting military operations, such as bomb disposal or surveillance in conflict zones.

However, robots have also raised several concerns. One concern is the displacement of humans from their jobs due to the increasing advancement and capability of robots, which can now perform tasks traditionally done by humans. According to a study by the McKinsey Global Institute, up to 800 million jobs could be lost to automation by 2030. This is equivalent to 14% of the global workforce.

Other concerns involve ethical implications. As robots become more independent, will they behave ethically and responsibly? Will they prioritize the best interests of humans or potentially cause them harm? Will they adhere to the ethical standards we hold ourselves to?

In 1942, Isaac Asimov, a science fiction author, introduced the three laws of robotics.

1. A robot may not injure a human being or, through inaction, allow a human being to come to harm.
2. A robot must obey the orders given to it by human beings, except where such orders would conflict with the First Law.
3. A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.

Although these laws were introduced in science fiction publications, they are based on the ethical considerations that humans are concerned about.

Overall, it is important to carefully balance the benefits of robotics while addressing these concerns as we move towards a more automated future—a future in which humans interact with computers, machines, and robots.

SUMMARY

Human-Computer Interaction is a multidisciplinary field that focuses on the study of the design of computers and its impact on the interaction that occurs between humans and computers. HCI include computers and other information technologies, particularly those that have an embedded computer.

The Human-Computer Interface is the boundary across which the two systems interact. For the computer, this includes all its input and output devices. For the human, this includes the sensory organs and hands. Interaction occurs in an input/output loop that results across the interface.

The interface of the computer can be expanded by installing peripheral devices. These devices are input and output devices that augment an existing functionality or add a new one. Peripherals enhance human-computer interaction.

Embedded computers have made a wide variety of machines and electronic devices smart. Introducing embedded computers enhances the human-machine interface by adding new functionalities and improving usability.

Robots are machines capable of performing tasks autonomously or with some human intervention. They are categorized depending on their autonomy, mobility and purpose. Robots bring the benefits of taking over menial and repetitive jobs, as well as dangerous ones. The concerns are that robots are taking over too many jobs leaving people jobless, and that they may not in the future act ethically and in our best interest.

Career Tips

Mechatronics is an interdisciplinary field that combines mechanical engineering, electrical engineering, and computer science. Mechatronic systems can be found in various applications, such as manufacturing, transportation, healthcare and robotics.

Activity 1.1

Read the following statements, and write on the line provided for each statement 'T' if it is true and 'F' if it is false.

T/F	STATEMENT
<u>T</u>	Humans interact with computers across an interface.
<u>F</u>	Interaction occurs in one direction, from human to computer.
<u>T</u>	The I/O loop is a cycle of exchanges of signals.
<u>F</u>	The computer's interface cannot be expanded.
<u>T</u>	Peripheral devices are input and output devices that can expand the interface.
<u>T</u>	Embedded computers make machines smart.
<u>T</u>	The smartphone used to be a cellphone with an embedded computer; now, it is a general-purpose computer.
<u>F</u>	All robots perform tasks autonomously.
<u>T</u>	Taking over menial jobs is considered a benefit of robots.
<u>F</u>	Isaac Asimov was a scientist.

Activity 1.2

Answer the questions below.

1. Which component is not part of the interface?
 - a) **Processor**
 - b) Mouse
 - c) Keyboard

2. Which device acts as a peripheral device for a tablet?
 - a) Touchscreen
 - b) Memory
 - c) **Digital pen**

3. Which device acts as a peripheral device for a laptop?
 - a) Screen
 - b) **Printer**
 - c) Keyboard

4. Which device acts as a peripheral device for a smartphone?
 - a) **Wireless earphones**
 - b) Touchscreen
 - c) Processor

5. Which device acts as an I/O device?
 - a) **Smartboard**
 - b) Keyboard
 - c) Whiteboard

Activity 1.3

Complete the sentences below.

1. The purpose of an interface in a machine is _____.
 - a) to control the machine's operations
 - b) **to provide functionality and usability**
 - c) to connect the machine to the internet

2. For a microwave oven, an embedded computer adds the functionality of _____.
 - a) cooking food
 - b) heating beverages
 - c) **programmable cooking modes**

3. For an air conditioner, an embedded computer adds the functionality of _____.
 - a) **automatically adjusting room temperature**
 - b) blowing hot air to warm up a room
 - c) blowing cold air to cool down a room

4. An example of a modern human-machine interface is _____.
 - a) dials and knobs
 - b) switches and buttons
 - c) **graphical user interface**

5. The first to have an embedded computer was the _____.
 - a) **Nokia 9000 Communicator**
 - b) Motorola DynaTAC 8000X
 - c) Apple iPhone

Activity 1.4

Answer the questions, and complete the sentences below.

1. Which of the following is an example of a task that a robot can perform in a factory?
 - a) Conducting underwater inspections
 - b) Exploring hostile environments
 - c) **Tightening screws on an assembly line**

2. Which category of robot can make decisions on its own without human intervention?
 - a) Semi-autonomous robot
 - b) Non-autonomous robot
 - c) **Autonomous robot**

3. A robotic vacuum cleaner that does not require human control is best described as _____.
 - a) stationary and non-autonomous
 - b) stationary and autonomous
 - c) **mobile and autonomous**

4. A robotic arm on an assembly line that requires some human control is best described as _____.
 - a) stationary and non-autonomous
 - b) **stationary and semi-autonomous**
 - c) mobile and autonomous

5. What is one potential benefit of using robots in hazardous environments?
 - a) Improved job opportunity
 - b) **Improved safety**
 - c) Improved cost savings

Activity 1.5

Answer the questions below.

1. Which types of computers do you use? List the devices that constitute each computer's interface.

Desktop: screen, keyboard, mouse, speaker, microphone, camera

Laptop: screen, keyboard, touchpad, speaker, microphone, camera

Tablet / smartphone: touchscreen, speaker, microphone, camera

2. What peripheral devices have you added to your computer's interface?

Wireless / wired earphone +\- microphone, webcam, printer, scanner

3. Of the peripheral devices you use, what functionality does each device provide you? How do you find that beneficial?

The printer allows me to print homework from the comfort of my home.

Earphones allow me to listen at higher volumes without bothering others.

Activity 1.6

Write your answer(s) on the lines provided below.

1. List smart machines found at home or school.

Smart (interactive) board

Smart thermostat or air conditioning unit

Smart TV

2. Most modern elevators have embedded computers that perform various tasks to ensure the safe and smooth operation of the elevator. Describe the user interface of a modern elevator.

Outside the elevator, there is a screen to show the floor number at which the cabin is currently at, as well as call buttons to call for the cabin.

Inside the elevator, there is also a screen to indicate the current floor, a control panel of floor numbers and a speaker to sound arrival at the requested floor.

3. “A smartphone is a computer, not a cellphone with an embedded computer.” Is this statement true? Explain.

This statement is mostly true as it depends on the main use or purpose of the phone. In general, the smartphone is mainly used for its general-purpose capabilities: navigating the web and social media, communicating via text or audiovisual messages, reviewing and completing assignments, attending online classes, playing games, listening to music, watching videos, etc.

Activity 1.7

Write your answer(s) on the lines provided below.

1. List three tasks robotic arms perform in factories.

Load and unload items

Weld and solder

Insert and tighten screws

2. Investigate the rovers sent to explore the surface of Mars. Choose one and list three of the various functions it can perform based on its sensors and actuators.

Curiosity landed on Mars in August 2012.

It has six wheel actuators that can move across rocky terrain.

It has sensors to monitor temperature, pressure, wind speed and radiation.

It has a robotic arm actuator to collect samples.

3. “Cars that can drive themselves are robots.” Is this statement true? Explain. .

If we define robots as machines capable of performing tasks autonomously

or under varying degrees of control, then self-driving cars can be considered

robots. These cars have sensors, actuators and onboard computers to perceive

the environment, make decisions and control their movements. They can

operate without direct human intervention.

LESSON 2: COMPUTER INTERACTION

Computers interact with each other by sending and receiving information just like their internal components. Information exchange follows similar principles as those that apply inside the computer. It, therefore, follows that there must be signals, a medium through which signals will travel, a way to organize these transmissions and addresses that indicate source and destination of transmitted information.

NETWORKS

A network is a system of interconnected computers and devices that are capable of exchanging information. Networks vary depending on their size, area of coverage, and the technology used to transmit information. The smallest network may consist of two devices in one room, while the largest network is the internet a system of communication that consists of billions of devices and spans the entire globe.

The method of information transfer depends on the technology used. In some cases, transmission occurs through wired connections where data is transmitted as electrical or light signals across wires and cables. Conversely, there is wireless transmission, which uses radio waves to send signals through the air.

A NETWORK IS A SYSTEM OF INTERCONNECTED COMPUTERS AND DEVICES CAPABLE OF EXCHANGING INFORMATION.

PERSONAL AREA NETWORK

The smallest network is a personal area network, PAN. It typically consists of a personal computer connected to peripherals and other devices in its vicinity. Any personal computer can be used to establish a PAN.

Typically, the PAN is centered around an individual and the personal computer being used. Connections in a PAN can be wired, wireless or a combination of both.

A PERSONAL AREA NETWORK IS A NETWORK CENTERED AROUND AN INDIVIDUAL.

EXAMPLE 2.1

Sarah sat in her school's library. She had some homework she needed to finish and print. She was using her tablet, which was connected to a digital pen, her earbuds and the library's printer. Sarah's devices were connected in PAN through wireless connections.

These short-range wireless connections are based on Bluetooth technology. Each device has a Bluetooth antenna capable of sending and receiving radio wave signals. Most Bluetooth antennas are only capable of a range of about 10 meters, they are referred to as Class 2 Bluetooth antennas and are a good choice for short-range communications. On the other hand, Class 1 Bluetooth antennas are capable of transmissions up to a range of 100 meters.

EXAMPLE 2.2

The librarian also had her own PAN. But in her case, the PAN was wired rather than wireless. Her desktop was connected to a printer, scanner, and a smartboard by cables. This allowed her desktop to exchange information with devices in its vicinity.

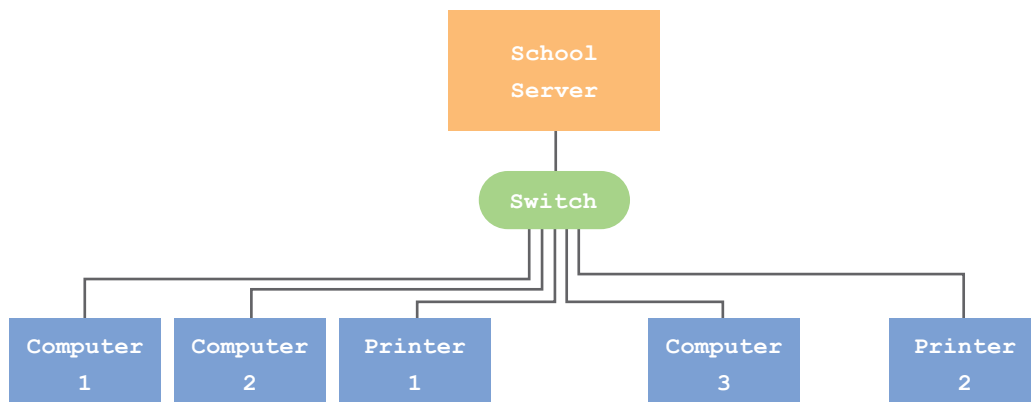
LOCAL AREA NETWORK

A local area network, or LAN, is larger than a PAN and covers a wider area such as a home, office building, or school campus. A LAN is more extensive than a PAN and serves a larger group of users.

EXAMPLE 2.3

The librarian can use her desktop to connect to other computers at school. This includes the school server which holds the school's database and information management system. By having access to the server, the librarian gains information about the inventory of books in the library, which books have been checked out and by who, as well as overdue to books. This information is shared with the school principal and teachers. Even students can login to view their list of checked out and overdue books.

Our hypothetical school LAN consists of a central server and all the computers connected to it at school. Connections are established by wires and cables, specifically Ethernet cables. This is a wired LAN, since all its connections are in the form of cables.



The figure shows the school as one local area network that consists of 3 computers, 2 printers and a server all of which connect and communicate via one switch.

Switches are networking devices that connect multiple computers and other devices within a network. They receive information from one device and forward it to the appropriate destination within a LAN.

A SWITCH IS A NETWORK DEVICE THAT CONNECTS MULTIPLE COMPUTERS AND DEVICES WITHIN A LAN.

WIDE AREA NETWORK

Wide area networks, WAN, are much larger. They vary in size of coverage from city-wide to as wide as our whole planet. The internet is the largest and most widely known example of a WAN. It is a global network of interconnected smaller networks that spans across the entire planet.

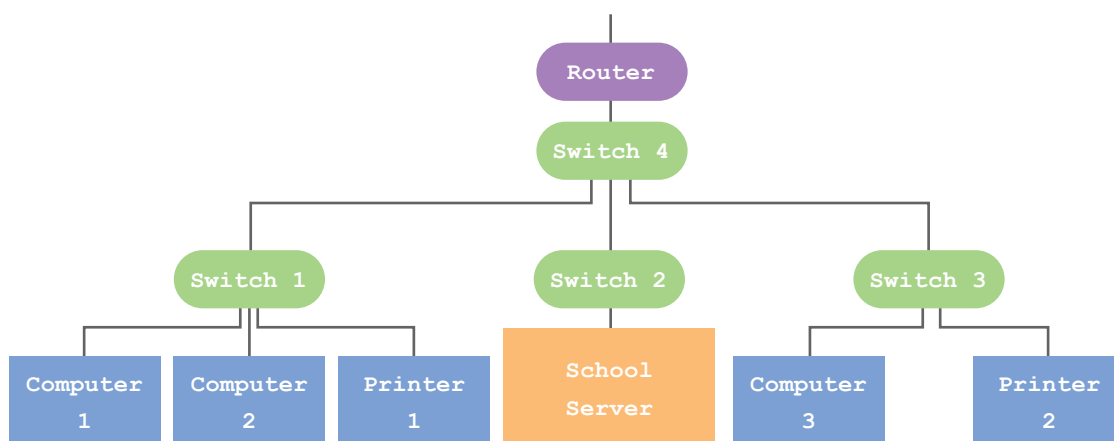
While a switch is capable of handle connections inside a LAN, it cannot perform the tasks necessary to handle connections outside its LAN. For such tasks, a router is needed. Routers act as gateways to the internet and its traffic controllers. When information has to pass outside a LAN, the router identify its destination and the best path to get there.

A ROUTER IS A NETWORK DEVICE THAT ACTS AS A GATEWAY TO THE INTERNET AND AS ITS TRAFFIC CONTROLLER.

EXAMPLE 2.4

In the previous example, we assumed a small school. A larger school would have many more computers, printers and other devices. Therefore, instead of having a large LAN, it is more appropriate to divide the school into smaller LANs.

Each section of the school would have its own switch and form its own LAN. To connect these smaller LANs, another switch would be needed forming one larger LAN. The addition of a router provides the school with access to the internet.



In this figure, the whole school campus is depicted as 1 larger LAN consisting of 3 smaller LANs. Each smaller LAN is established by a switch. All three switches are connected by Switch 4. The router provides connectivity to the internet.

The internet is considered to be the widest network. The backbone of the internet is a complex series of submarine fiber optic cables that connect all the continents and transmit information in the form of light signals. As of early 2023, there are 552 active and planned submarine fiber optic cables used for the internet. These cables span over 1.4 million kilometers and carry more than 95% of the world's international internet traffic.

THE INTERNET IS A COMPLEX SYSTEM OF NETWORKS WORKING TOGETHER TO PROVIDE COMMUNICATION ACROSS THE GLOBE.

But the internet does not solely rely on fiber optic cables for signal transmission, it also employs satellite and telephone communication networks, such as land lines and the cellphone grid.

ADAPTERS

The three common forms of energy used to as communication signals are electrical, light and radio waves. Electrical signals and light signals are used to transmit information in wires, typically copper wires for electrical signals and fiber optic cables for light, while radio signals are used to transmit information through the air. Each type of signal requires a specific type of hardware to handle its conversion but in general, communication channels are broadly categorized as wired and wireless.

THERE ARE TWO BROAD CATEGORIES OF COMMUNICATION CHANNELS WIRED AND WIRELESS.

Since the signals of communication traveling in a network are different from the electronic digital signals of the computer, signals must be converted. This is the task of the network adapter, a hardware component of the computer interface. The job of the adapter is to convert the internal signals of the computer to the external signals of the network and vice versa. As such, the adapter is considered an I/O device.

EACH ADAPTER CONVERTS INTERNAL SIGNALS TO A SPECIFIC TYPE OF EXTERNAL SIGNAL INTENDED FOR TRANSMISSION.

Wired Adapters

Each of the adapters converts electronic digital signals into a different type of transmission signal. They typically connect via a port.

1. **Land line Modems** convert between the electronic digital signals of the

computer and the continuous (analog) electrical signals used to transfer information through telephone land lines. They connect LANs to the internet.

2. **Ethernet network interface card, NIC**, converts between the electronic digital signals of the computer and another electronic digital signals used to transfer information through long cables. It also connects devices in LANs and the connect LANs to the internet.
3. **Fiber optic NIC** typically includes a fiber optic transceiver, which converts the electrical signals used by the device into optical signals that can be transmitted over the fiber optic cable. It is commonly used in data centers where high-speed data transmission and reliability are critical.

Wireless Adapters

The following three adapters convert between the electronic digital signals of the computer and radio waves, which are used to transfer information through the air. They capture and emit signals using an antenna. One main difference between them is their range of transmission.

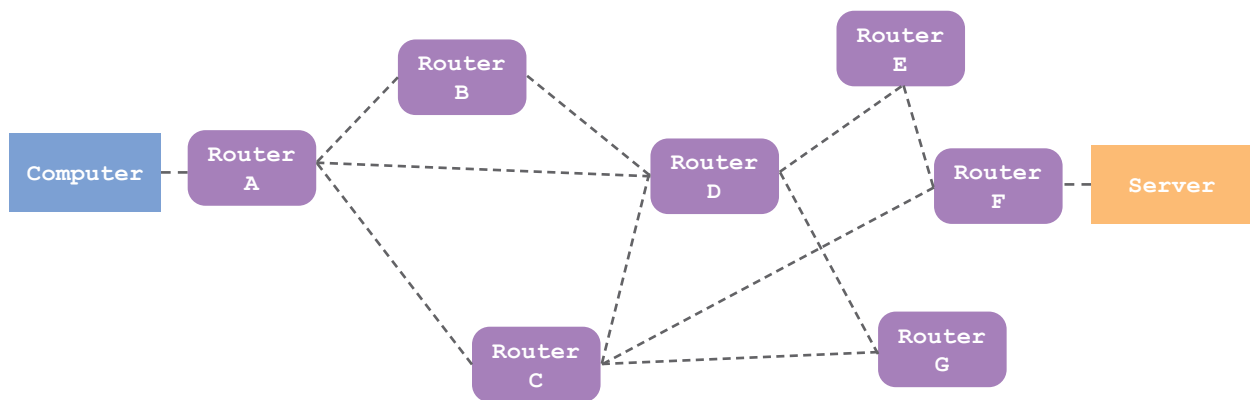
1. **Cellular Modem** is included as a component in smartphones and some tablets and is designed to transmit signals over several kilometers and is used to connect to the internet via cellular networks.
2. **WiFi NIC** can transmit signals up to 45 meters indoors and up to 150 meters outdoors. It is typically included as an internal component in most personal computers or can be externally connected to one. It is used to connect to LANs and the internet.
3. **Bluetooth Adapter** can transmit special type of radio wave signal, referred to as frequency-hopping spread spectrum (FHSS) used for short distance of upto 10 meters indoors and is used to connect to nearby devices. This allows the computer to established a personal area network between devices in close proximity such as Bluetooth-enabled earphones, headphones, speakers, television sets and other personal computers.

BEST ROUTE

The internet is a busy place. Once the signal leaves the source computer, depending on the distance it must travel, it will likely have to pass through several routers before arriving at its destination. The path the signal takes from source to destination is determined by routers.

EXAMPLE 2.4

A computer sends a request to a server. Router A is acting as the computer's gateway to the internet. Therefore, Router A needs to know the best route to send the signal to the server.



One method of determining the best route for delivering a signal to its destination is to count the number of routers the signal must pass through before arriving. This is referred to as the hop count. Every router keeps a routing database of all the possible routes and their metrics, such as the hop count.

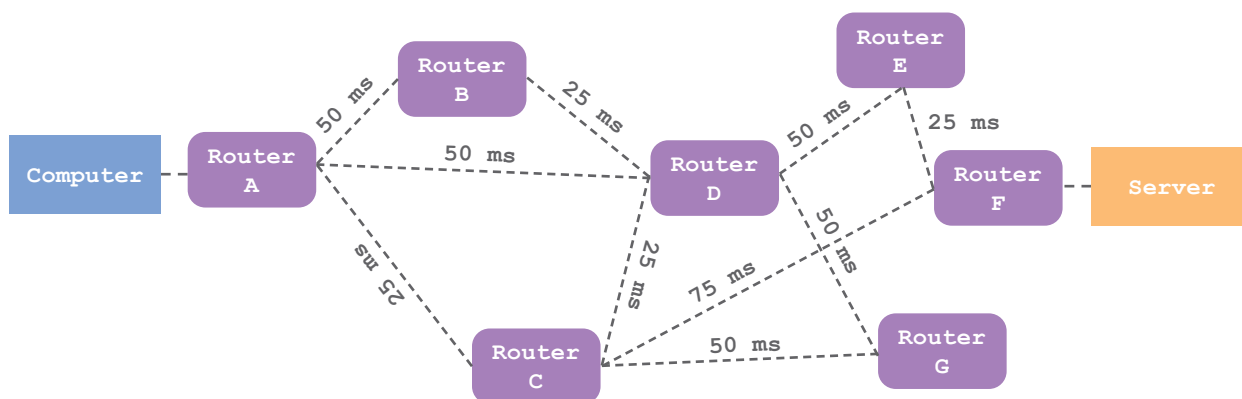
Based on this metric, Router A would take route option 3 because it has the fewest hops from source to destination.

Route Option	Hop 1	Hop 2	Hop 3	Hop 4	Hop Count
1	B	D	E	F	4
2	D	E	F		3
3	C	F			2

Another metric routers can use for determining the best paths is latency, or the time it takes for a signal to travel from one point in the network to another. One common method for measuring latency is the ‘ping’, which measures the round-trip time for a signal to travel from the source to the destination and back.

In some network configurations, routers may ping their neighboring routers. These latency measurements are then shared with other routers, which use them to choose the best route for sending signals.

A long ping indicates either that the distance between the source and destination is long or that there are delays in processing the signal, due to either heavy traffic or a slow router.



The figure shows the latency time in milliseconds between routers. Router A would require such information to determine the best route to its destination, the server. Total latency is estimated as the sum of all intermediary latencies.

Route Option	Hop 1	Hop 2	Hop 3	Hop 4	Total Latency
1	B: 50	D: 25	E: 50	F: 25	150 ms
2	D: 50	E: 50	F: 25		125 ms
3	C: 25	F: 75			100 ms

According to the database information, route option 3 remains the best route to the destination. This route not only has the fewest hops, 2, but also has the shortest latency, 100 ms.

THE BEST PATH IS DETERMINED USING METRICS SUCH AS HOP COUNT AND LATENCY.

What if Router C becomes congested with many requests to route signals through? What if the latency between Router A and Router C becomes 100 ms? In such a case, route option 3 would not seem to be so desirable having a total latency of 175 ms, the highest among all three options. Instead, route option 2 becomes the best route to the destination.

Some cases are not as clear. What if the latency between Router A and Router C becomes 50 ms? Now route options 2 and 3 have the same total latency, 125 ms.

In general, routers differ based on the decision-making program used to determine the best route. Some programs place greater importance on hop count, while others favor latency, and still others may use yet another metric.

TRUST

When transmitting signals across the internet, or between any two devices, trust must be established. The user must be able to trust that the signals are transmitted without any unintentional error and without any intentional change or theft. The transmission must maintain the integrity and security of the message it is carrying.

**THE USER TRUSTS THAT THE TRANSMISSION WILL MAINTAIN THE
INTEGRITY AND SECURITY OF THE MESSAGE.**

ERRORS

Errors in data during transmission can happen for many reasons, which, for the most part, are generally related to the phenomena of attenuation and interference.

Loss of strength, or attenuation, can occur with any form of signal in any medium if that signal is required to travel great distances. The longer the distance, the more likely it is that the signal will lose its strength, whether it is light in a fiber optic cable, electricity in a copper wire, or radio waves traveling through air.

**ERRORS IN DATA TRANSMISSION HAPPEN GENERALLY DUE TO
ATTENUATION OR INTERFERENCE.**

Another common cause of errors is interference. There are several reasons for interference, such as storms, buildings, neighboring wires, and cosmic and solar radiation.

Storms and severe weather can interfere with and disrupt radio waves, while obstacles such as buildings can block or reflect radio waves, creating multiple paths of waves that interfere with each other.

Nearby electrical wires or electronic devices can generate electromagnetic radiation that interferes with the signals, particularly electrical signals in copper wires.

Other forms of energy, such as cosmic and solar radiation, can be problematic for satellites in low earth orbit and other devices, such as computers and routers located at high altitudes.

PROTECTION

There are methods used to protect against errors and decrease the risk of their occurrence. For signal attenuation, repeaters are placed at a certain distance along the path of propagation of the signal, whether it is light in fiber-optic cables, electricity in copper wires, or radio waves traveling through the air. Repeaters are devices that capture, amplify, and retransmit the same signal, so signals whose energy is weakening become restrengthened.

Shielding works to protect against interference in copper wires and satellites. For copper wires, adding an insulating layer around the wires helps shield the signals inside from outer, neighboring interference. As for satellites, various types of shielding and design techniques are used to protect the internal electronic circuits from cosmic, and to some extent, solar radiation.

REDUNDANCY IN DATA MEANS MAINTAINING ADDITIONAL COPIES OF THE ORIGINAL DATA.

A simple and common strategy used to protect from complete loss of information is redundancy, a term which means having more of the same. When any transmission occurs, whether it is between components of a computer or between computers, only a copy of the original information is sent. The original information, or file, stays put. It is copied, and that copy is the signal that is transmitted. If the transmitted information is lost or disrupted, the original information is still safe and can be recopied and retransmitted.

DETECTION

When a signal is transmitted, the receiving computer sends back an “All is Good” signal to the transmitting computer, which basically means that “I have received your message.” This is referred to as an acknowledgement. If the acknowledgement is not received, the sender will assume that the message was lost and will transmit another copy.

If the message does arrive, then it must be checked for errors. There are several methods of error checking, one common method is a checksum. A checksum is a value calculated from the data that is sent along with the data itself. The receiver can calculate its own checksum from the received data and compare it to the one that was sent. If they match, the data is correct; if not, an error occurred.

THE ACKNOWLEDGEMENT AND THE CHECKSUM ARE TWO METHODS OF DETECTING ERRORS IN DATA TRANSMISSION.

EXAMPLE 2.5

The checksum works at the byte level. Let's assume that the message "Hello" is going to be sent. Every character in the alphabet has an 8-bit binary code to represent it. This byte actually represents the number assigned to the character.

H	e	l	l	o	
0100 1000	0110 0101	0110 1100	0110 1100	0110 1111	
72	101	108	108	111	
Checksum	72 + 101 + 108 + 108 + 111 =				500

In this case, the values for each of the 5 characters in the word "Hello" are added. The sum, referred to as the checksum, is sent along with the message. The computer on the receiving end will repeat the checksum calculation on the message it received. If the receiver's checksum is equal to the sender's checksum then the signals are all correct and no errors have occurred during transmission. The message the receiver has is the same as the one sent by the sender.

ENCRYPTION

Just like we expect to maintain the integrity of a message during transmission, we also expect that the transmission will be secure. The message should arrive error-free and it should not be amenable to hacking. Hacking means gaining unauthorized access to a computer, network or transmission, with the intention of changing, stealing or destroying data.

HACKING IS GAINING UNAUTHORIZED ACCESS TO A COMPUTER, NETWORK OR TRANSMISSION, TO CHANGE, STEAL OR DESTROY INFORMATION.

One way to keep a message secure from hackers is to encrypt it. Encryption changes the message in such a way as only the sender and receiver can read it. So even if hackers gained access to the message they would not know what to do with it because they cannot read it.

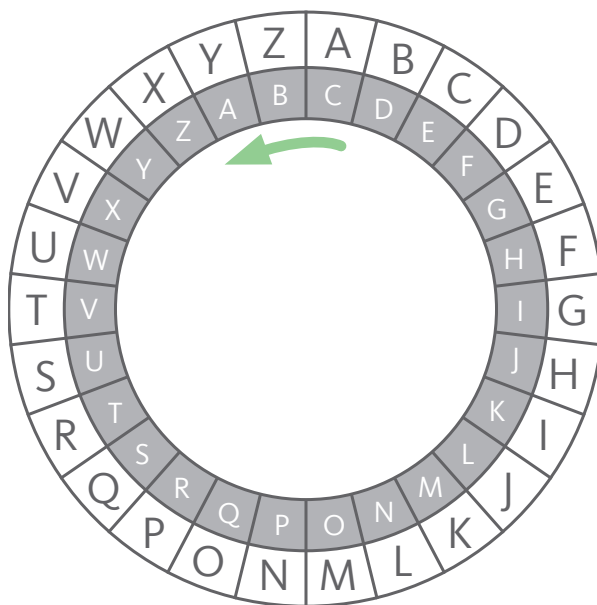
ENCRYPTION IS THE PROCESS OF REWRITING A MESSAGE IN A WAY THAT CAN ONLY BE READ BY A USER WHO HAS THE KEY.

Encryption is a process that transforms a message from a readable form, plaintext, into an unreadable form, ciphertext. The transformation is based on a set of steps, an algorithm, that determines how the message will be changed. The encryption key determines by how much the message will be changed.

EXAMPLE 2.6

The Caesar cipher is a simple encryption algorithm that dates back to the time of the Roman Empire. It is named after Julius Caesar, who is said to have used it to protect the confidentiality of his military messages. This allowed Caesar to send messages across the span of the empire with the security that should a message fall into the wrong hands it could not be read.

The Caesar cipher works by shifting each letter in the plaintext by a fixed number of positions in the alphabet. A Caesar cipher disk, is a tool used to encrypt and decrypt messages using the Caesar cipher. It consists of a circular disk or wheel with the alphabet printed on it, typically in two concentric rings. The outer ring represents the plaintext alphabet, while the inner ring represents the corresponding ciphertext alphabet after applying a specific shift value.



A Caesar cipher disk whose inner ring is turned 2 positions counterclockwise. 'A' on the outer ring corresponds to 'C' on the inner ring, 'B' corresponds to 'D', 'C' corresponds to 'E', and so on. To encrypt a message replace each letter of the plaintext on the outer ring with the corresponding letter of the ciphertext on the inner ring. In this case, the word "HELLO" would be encrypted as "JGNNQ".

The encryption algorithm would be to shift the inner wheel in a certain direction by a certain number of positions. The key would be the direction and number of positions to shift, which, in the case of this example, is 2 positions counterclockwise.

Encryption algorithms and keys are far more elaborate when considering computer-to-computer data transmission. Every communication application would have its own encryption algorithms and keys that it uses to ensure that all messages and information sent by it are secure.

Although this presents a very simplified view of computer-to-computer interaction, or data transmission, it does emphasize the importance of establishing trust by ensuring the integrity and security of transmitted information.

Career Tips

Cybersecurity is a rapidly evolving field that encompasses various aspects of protecting computer systems, networks, and data from unauthorized access, attacks, and damage.

Ethical hackers, also known as ‘white hat’ hackers are professionals who are authorized to identify vulnerabilities and weaknesses in computer systems, networks, or applications with the goal of strengthening their security measures.

SUMMARY

A network is a system of interconnected computers and devices capable of exchanging information. Networks vary depending on size, area of coverage, and the technology used to transmit information.

The smallest network is a personal area network, PAN, which is a network centered around an individual. Any personal computer can be used to set up a PAN. It can be wired, wireless or a combination of both.

A local area network, LAN, is larger than a PAN and covers a wider area and serves a larger number of users. A switch is a network device that connects multiple computers and devices within a LAN.

Wide area networks, WAN, vary in size and coverage. They consist of multiple LANs and can provide coverage for a city. The largest WAN is the internet. The internet is a complex system of networks working together to provide communication across the globe. A router is a network device that acts as a gateway to the internet and as its traffic controller.

The adapter is a computer component that converts the internal signals of the computer to the external signals of the network and vice versa. Each adapter converts internal signals to a specific type of external signal intended for transmission. There are two broad categories of communication channels wired and wireless. Wired adapters are: land line modems, ethernet NICs, and fiber optic NICs. Wireless adapters are: cellular modems, WiFi NICs, and Bluetooth adapters.

Routers are tasked with identifying the best route between source and destination. The best path is determined using metrics such as hop count and latency.

Transmission must maintain the integrity and security of the message. Errors in data transmission happen generally due to attenuation or interference. To protect against signal attenuation, repeaters are used to amplify, and retransmit signals whose energy is weakening. To protect against interference, shielding is used around copper wires and on satellites.

The acknowledgement and the checksum are two methods of detecting errors in data transmission. If the acknowledgement is not received, the sender will assume that the message was lost and will retransmit. If the receiver's checksum is not the same as that of the sender, then there is an error in the message.

Encryption is the process of rewriting a message in a way that can only be read by a user who has the key. Every encryption requires an encryption algorithm and key to transform the message from readable to unreadable forms.

Activity 2.1

Read the following statements and write on the line provided for each statement 'T' if it is true and 'F' if it is false.

T/F	STATEMENT
<u>F</u>	A network does not act as a system.
<u>T</u>	The smallest network is a personal area network.
<u>T</u>	The internet is a complex system of networks.
<u>T</u>	The adapter is considered an I/O device.
<u>F</u>	Only two forms of energy are used to convey signals: light and electricity.
<u>T</u>	Network interface cards and modems are different types of adapters.
<u>F</u>	The best route is determined based only on the hop count.
<u>F</u>	Errors in data transmission happen only due to attenuation.
<u>T</u>	The checksum is one method to detect errors.
<u>F</u>	A key is not required to encrypt a message.

Activity 2.2

Complete the sentences below.

1. A network is _____.
 - a) a device that stores information
 - b) a technology used for sending signals
 - c) a system of interconnected computers and devices capable of exchanging information

2. The internet transmits information _____.
 - a) only through wired connections
 - b) only through wireless transmissions
 - c) using a combination of both wired and wireless transmissions.

3. The primary function of a switch in a network is to _____.
 - a) serve as a power supply.
 - b) receive and forward signals within a LAN
 - c) connect a LAN to the internet

4. The device used to handle connections outside a LAN is _____.
 - a) a server
 - b) a switch
 - c) a router

5. The backbone of the internet is formed by _____.
 - a) a complex series of submarine fiber optic cables
 - b) satellite and telephone communication networks
 - c) routers and switches

Activity 2.3

Answer the questions below.

1. What are the three common forms of energy used for communication signals?
 - a) Magnetic, sound and light
 - b) **Electrical, light and radio waves**
 - c) Heat, mechanical and electrical

2. What are the two broad categories of communication channels?
 - a) Internal and external
 - b) Visible and invisible
 - c) **Wired and wireless**

3. What is the primary function of a network adapter?
 - a) **To convert between internal computer signals and external network signals.**
 - b) To convert AC power to DC power.
 - c) To increase the speed of the computer processor.

4. Land line Modems convert electronic digital signals to what kind of signals?
 - a) Radio waves
 - b) **Continuous (analog) electrical signals**
 - c) Frequency-hopping spread spectrum signals

5. Bluetooth Adapters can transmit a special type of radio wave signal referred to as?
 - a) **Frequency-hopping spread spectrum (FHSS)**
 - b) Continuous wave spectrum (CWS)
 - c) Modulated wave spectrum (MWS)

Activity 2.4

Complete the sentences below.

1. The role of a router in internet connectivity is _____.
 - a) to send signals directly from the source computer to the destination server.
 - b) to store data from the source computer.
 - c) to determine the path the signal takes from source to destination.

2. Every router keeps _____.
 - a) a record of all the users in the network.
 - b) a routing database of all the possible routes and their metrics.
 - c) a list of all the signals it has processed.

3. In networking, a ping is used for _____.
 - a) measuring the round-trip time for a signal to travel from the source to the destination and back
 - b) counting the number of routers the signal has passed through
 - c) identifying faults in the network

4. In networking, latency is the _____.
 - a) time it takes for a signal to go and come back
 - b) distance between source and destination
 - c) time it takes for a signal to arrive at its destination

5. In general, routing algorithms use _____.
 - a) different metrics
 - b) only hop count as a metric
 - c) the same metrics

Activity 2.5

Answer the questions below.

1. What can cause signal attenuation during transmission?
 - a) The strength of the device sending the signal
 - b) **The distance the signal has to travel**
 - c) The user's trust in the transmission system

2. What are some common causes of interference during signal transmission?
 - a) **Storms and neighboring wires**
 - b) Rainbows and moonlight
 - c) Animals and plants

3. What is the purpose of a repeater in signal transmission?
 - a) To create a copy of the signal
 - b) **To capture, amplify, and retransmit the same signal**
 - c) To protect the signal from interference

4. What is the purpose of an acknowledgement in signal transmission?
 - a) To inform the sender that the message was sent
 - b) **To inform the sender that the message was received**
 - c) To inform the sender that the message was encrypted

5. What is encryption used for in signal transmission?
 - a) To increase the speed of transmission
 - b) To reduce errors in transmission
 - c) **To protect the data from hackers**

Activity 2.6

Answer the questions below.

1. Give an example of a PAN you use in your daily life.

Answers will vary. Example: My smartphone connected to my Bluetooth headphones and smartwatch.

2. Give an example of a LAN you use in your daily life.

Answers will vary. Example: My computer science classroom has a switch that connects the desktops to each other and to the school server allowing them to share resources.

3. How do routers function in a network?

Routers act as gateways to the internet and control its traffic. When information has to pass outside a LAN, the router identifies its destination and the best path to get there.

Activity 2.7

Answer the questions below.

1. How is the function of a Fiber optic NIC different from an Ethernet NIC?

A Fiber optic NIC converts the electrical signals used by the device into optical signals that can be transmitted over the fiber optic cable, while an Ethernet NIC converts between the electronic digital signals of the computer and another electronic digital signals used to transfer information through long cables.

2. Why are Fiber optic NICs commonly used in data centers?

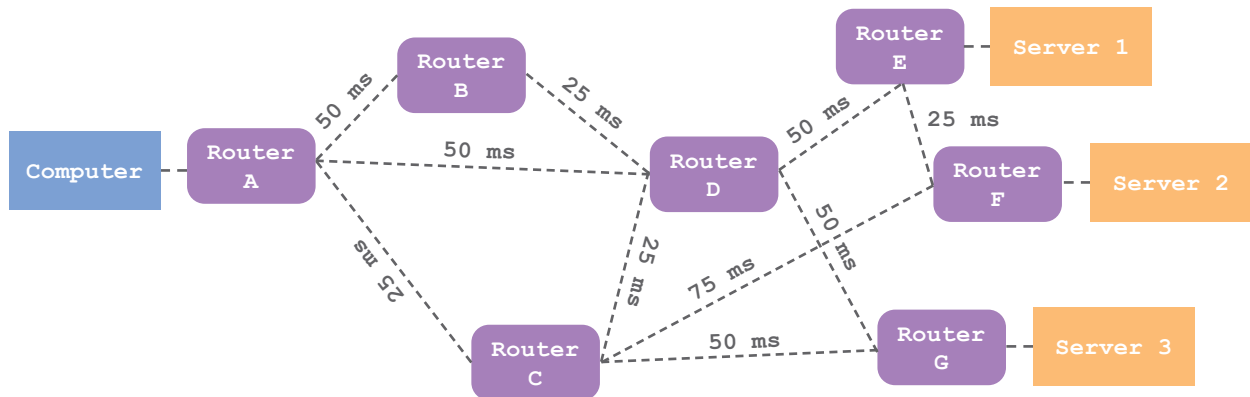
Fiber optic NICs are commonly used in data centers because they offer high-speed data transmission and reliability, which are critical in data center environments

3. Examine your personal computer and a desktop computer at school. List the various network adapters these computers have.

The answers will vary widely depending on the computers being examined.

Activity 2.8

Use the network diagram to answer the questions below.



1. Complete the routing database below between Computer and Server 1.

Route Option	Hop 1	Hop 2	Hop 3	Hop 4	Total Latency
1	B: 50	D: 25	E: 50		125 ms
2	D: 50	E: 50			100 ms
3	C: 25	D: 25	E: 50		100 ms
4	C: 25	F: 75	E: 25		125 ms
5	C: 25	G: 50	D: 50	E: 50	175 ms

2. Complete the routing database below between Computer and Server 2.

Route Option	Hop 1	Hop 2	Hop 3	Hop 4	Total Latency
6	B: 50	D: 25	E: 50	F: 25	150 ms
7	D: 50	E: 50	F: 25		125 ms
8	C: 25	D: 25	E: 50	F: 25	125 ms
9	C: 25	F: 75			100 ms

3. Complete the routing database below between Computer and Server 3.

Route Option	Hop 1	Hop 2	Hop 3	Hop 4	Total Latency
10	B: 50	D: 25	G: 50		125 ms
11	D: 50	G: 50			100 ms
12	C: 25	G: 50			75 ms

4. Based only on hop counts, list the best route to each server. Use latency to resolve a tie.

Computer to Server 1: Route Option 2

Computer to Server 2: Route Option 9

Computer to Server 3: Route Option 11 and 12, based on latency 12 wins.

5. Based only on latency, list the best route to each server. Use hop count to resolve a tie.

Computer to Server 1: Route Option 2 and 3, based on hop count 2 wins.

Computer to Server 2: Route Option 9

Computer to Server 3: Route Option 12

6. Use Command Prompt or Terminal to measure ping for the web address, www.google.com, and any other of your choosing.

ping <web address>

Activity 2.9

Answer the questions below. Use an ASCII table to identify the decimal number that represents each character. Note that capital and small characters have different ASCII values and that a “space” has a value.

1. You send a message, “Hi there”, to your friend. The checksum at the recipient’s end is 704. Did the message arrive intact without error?

The checksum, based on ASCII values, is 745.

So there is an error in the message.

2. You send a message, “I finished homework”, to your friend. The checksum at the recipient’s end is 1855. Did the message arrive intact without error?

The checksum, based on ASCII values, is 1855.

So there is no error in the message.

3. A transaction code is sent between two banks, “AA103XT47”. The checksum at the recipient’s end is 551. Did the message arrive intact without error?

The checksum, based on ASCII values, is 557.

So there is no error in the message.

Activity 2.10

The program below, in Python, is intended to accept a message from the user and to calculate the checksum. Use this program to check your answers for Activity 2.9.

```
def calculate_checksum(message):
```

```
    checksum = 0
```

```
    for char in message:
```

```
        ascii_value = ord(char)
```

```
        checksum = checksum + ascii_value
```

```
    return checksum
```

```
message = input('Enter message: ')
```

```
checksum = calculate_checksum(message)
```

```
print (f'The checksum for message is: {checksum}')
```

Defines the function that calculates the checksum by adding the ASCII values.

Defines the checksum as a variable and sets it to zero.

Starts a loop that takes a character from the message each time.

Converts character to its ASCII value.

Updates checksum by adding the new ASCII value to the current checksum value.

Repeats the loop.

Gets a string input from the user and assigns it to "message".

Calculates the checksum using the function "calculate_checksum".

Prints the checksum.

Notes

The intention of this activity is to familiarize students with coding and the concept of algorithms and iteration. The intention is not to teach Python or any other coding language. Students should, however, appreciate and understand each step in the line of code.

Activity 2.11

Use a Caesar cipher wheel to answer the questions below. You may choose to construct one in class.

1. Encrypt the message “Hi there”. The key = counterclockwise 2.

Fg rfcpc

2. Encrypt the message “Beware of hackers”. The key = clockwise 5.

Gjbfwj tk mfhpjwx

3. Encrypt the message “Beware of hackers”. The key = counterclockwise 10.

Rumqhu ev xqsauhi

Activity 2.12

The program below, in Python, is intended to accept a message from the user and to calculate the checksum. Use this program to check your answers for Activity 2.9. Note that this program only performs a clockwise shift. To perform a counterclockwise shift subtract the shift value instead of adding it.

```
def caesar_cipher_encrypt(plaintext, shift):

    Ciphertext = " "

    for char in plaintext:

        if char.isalpha():
            if char.isupper():
                encrypted_char = chr((ord(char) - 65
                    + shift) % 26 + 65)

            else:
                encrypted_char = chr((ord(char) - 97
                    + shift) % 26 + 97)

            ciphertext = ciphertext + encrypted_char

        else:
            ciphertext = ciphertext + char

    return ciphertext

plaintext = input("Enter message: ")

shift = int(input("Enter shift value: "))

encrypted_message = caesar_cipher_encrypt(
    plaintext, shift)

print("Encrypted message: ", encrypted_message)
```

Defines the function that changes original plaintext to ciphertext by a specified shift.

Defines the ciphertext as an empty string.

Starts a loop that takes a character from the message each time.

Checks if the character is alphabetic.

Checks if the character is upper case.

If the character is uppercase, it calculates the ASCII value of the encrypted character by subtracting 65 (the ASCII value of 'A') from the ASCII value of the current character. The shift value is added, and the modulo 26 operation is applied to ensure the result stays within the range of the alphabet. Finally, 65 is added back to obtain the ASCII value of the encrypted character. The chr() function converts the ASCII value back to a character.

If the character is not uppercase, the same process is applied, but with 97 (the ASCII value of 'a') as the base value.

The encrypted character is added to the ciphertext string.

If the character is not alphabetic it is added as is to the ciphertext string.

Repeats the loop.

Gets a string input from the user and assigns it to "message".

Calculates the checksum using the function "calculate_checksum".

Prints the checksum.

LESSON 3: CLOUDS & THINGS

The basic model for a computer was a device that provided the four basic functions of input, output, memory and processing. However, this model has gradually evolved, and the four basic functions decoupled and dispersed across a wide geographical area. No longer do these four basic functions have to occur on one device.

The internet brought about two advances in computing models, the cloud and the internet of things.

THE INTERNET BROUGHT ABOUT THE CLOUD AND THE INTERNET OF THINGS.

THE CLOUD

The term “cloud” refers to a system of servers that are accessed over the Internet. The term originated from the cloud-shaped symbol often used in diagrams to represent the internet.



The ability to connect to the internet and share information with other computers brought about possibilities. What if these computers could store a whole lot more of your information? What if these computers could provide additional computational power and could process information that could not be performed by your computer?

This additional storage and processing power would greatly enhance the functionality of your computer. Storage does not need to be on your device,

it can be anywhere as long as you can access it whenever you want and from wherever you want.

STORAGE CAN BE OFFERED AS A CLOUD-BASED SERVICE.

EXAMPLE 3.1

Mario likes taking photos. All the photos he takes are stored in the drive of his smartphone. But the more photos he takes the less space he has available. What will happen when one day the drive is full? In such a case, Mario needs more storage space. He has three options:

1. **Memory Card:** If Mario's smartphone has a slot for a memory card, he could consider buying one to expand the storage. However, not all smartphones have this option, and even if his does, there's a limit to how much a memory card can store. Moreover, if Mario loses or damages his phone, he could potentially lose all his photos.
2. **External Drive:** External drives like flash drives, hard disk drives (HDD), or solid state drives (SSD) are also viable options. They provide ample storage and can be a good choice if Mario prefers having physical control over his data. The downside is they can be lost or damaged. Additionally, accessing or sharing the data might be inconvenient as Mario will have to carry the external drive around.
3. **Cloud-Based Storage Service:** Google Drive and Apple iCloud are examples of services that provide additional storage space for users and allow them to access files from any device that has internet access. They also automatically backup data, so if Mario loses his phone, his photos will still be safe. The downside is that if he doesn't have an internet connection, accessing his photos could be difficult. Also, storing data on the cloud means trusting the provider with the privacy and security of his data.

Storage Option	Advantages	Disadvantages	Maximum Storage Capacity	Approximate Price per GB
Memory Card (microSD card)	Portable and expandable storage. Immediate access to data.	Limited by phone compatibility and capacity. Risk of physical damage or loss.	Up to 1TB	Around \$0.20 - \$0.50
External Drive (HDD, SSD)	Large storage capacity. Data under physical control.	Risk of physical damage or loss. Not as portable or convenient as other options.	HDD: Up to 16TB SSD: Up to 8TB	HDD: Around \$0.02 - \$0.03, SSD: Around \$0.10 - \$0.20
Cloud Service (Google Drive, Apple iCloud)	Accessible from any device with internet. Automatic back-up and syncing.	Requires internet access. Recurring cost for extra storage. Dependence on service provider for data security.	For individual users, up to 2TB	Around \$0.025 - \$0.05

As for processing, that too can be provided as a cloud-based service. If you are connected to a much larger and more powerful computer then you could perform far more complex tasks than what your computer could perform alone.

PROCESSING CAN BE OFFERED AS A CLOUD-BASED SERVICE.

EXAMPLE 3.2

Liam loves playing video games. He has a basic laptop that he uses for school-work, but it's not powerful enough to run the latest high-definition games that require powerful processors and high-end graphics cards.

Google Stadia is a cloud-based gaming service. It allows Liam to play his favorite games directly from the cloud. The games are run on powerful servers,

and the video is streamed to Liam's laptop. His interactions, like button presses and mouse clicks, are sent back to the server allowing him to play more complex games than what he could on his computer.

A DATA CENTER IS A FACILITY THAT HOUSES SERVERS WORKING AS AN INTEGRATED SYSTEM SERVING THE REQUESTS OF MANY USERS.

DATA CENTERS

A datacenter is a facility that contains a great number of servers. The number of servers in a data center varies from tens to thousands depending on the size of the facility. It is important to realize that a data center is not simply a room filled with servers, it is a system of integrated components, such as servers, switches, power units, cooling systems and security systems, that come together to provide services such as the cloud-based services discussed in the previous section.

SERVERS

The heart of a data center is the server, a powerful computer designed to handle large amounts of data and run applications that serve other computers and their users across a network.

Servers are typically mounted on top of each other in metal frames called racks. Racks not only house the servers but also the other hardware components of the data center, such as the cables and switches that are used to connect servers together. In fact, a data center is one big, complex network consisting of servers connected by cables and switches.

Servers, like any other computer, have a basic architecture that also consists of the four basic components: input, output, memory and processing.

Although at certain terminals in a datacenter, there are keyboards, mice and

screens for data center administrators to take care of the internal workings of the servers, a server's main input and output component is a network adapter. This is because the majority of a server's input comes from the internet or the network on which it serves. Remember that the server is serving users by performing the tasks they request. As such, the server's input comes from the users and its output must be sent back to the users. So, whether it is input or output, the majority is being transmitted across a network, making the network adapter the most important I/O device for a server.

As for memory and processing components, servers also have central processing units (CPU), graphics processing units (GPU), caches, main memory and storage components.

PERFORMANCE

A data center is designed to respond to the requests of many users concurrently. Therefore, the performance of a server is measured by its throughput, which is defined as the number of tasks it can complete in 1 second.

A single server can handle thousands to tens of thousands of requests per second under regular conditions. Across an entire data center, the cumulative throughput could be in the millions or even billions of requests per second.

THROUGHPUT OF A SERVER IS THE NUMBER OF REQUESTS PROCESSED PER SECOND.

Throughput is dependent on the amount of data a server can receive in order to process, the more data it can process in one second the higher is its throughput. The amount of data that can be sent or received is referred to as the bandwidth which is measured as bits per second (bps), and ranges from megabits (Mbps) to gigabits (Gbps) and even to terabits (Tbps) per second.

Adequate bandwidth is critical to ensure that data can be transmitted rapidly,

enabling high-speed communication between servers and users or between different parts of the network.

Both throughput and bandwidth are interdependent. A server can have high throughput, but if the bandwidth is insufficient, data delivery to the server would be too slow, and the server won't be able to process tasks at its full capacity. On the other hand, a network with high bandwidth won't be fully utilized if the server's throughput is low.

BANDWIDTH IS THE AMOUNT OF BITS TRANSMITTED PER SECOND AND CAN LIMIT THE PERFORMANCE OF A DATA CENTER.

The performance of a server, its throughput, is enhanced by implementing parallel processing, which is the execution of multiple tasks at the same time. This method has already been implemented in the multi-core architecture of a CPU. Having multiple cores enhances the performance of a CPU by allowing it to process multiple tasks in parallel.

In the context of a data center, servers are arranged in clusters. This internal network organization allows the implementation of parallel processing at the level of the server. Multiple interconnected servers in a cluster can work simultaneously to perform multiple tasks for multiple users.

SERVERS ARE ARRANGED IN CLUSTERS THAT ALLOW THE PARALLEL PROCESSING OF TASKS.

Notes

Concurrent processing is processing by rapidly switching between tasks. Parallel processing is when processing of multiple tasks occurs simultaneously. Both enhance performance.

Data centers have two bandwidths, the internal and external bandwidths. This is because servers communicate with one another inside the data center via an internal network, and they also communicate with their clients via the network that connects them to the outside world.

DATA CENTERS HAVE BOTH AN INTERNAL AND EXTERNAL BANDWIDTHS.

If servers in a data center are working together then the internal bandwidth on the network interconnecting them must be high enough to not limit their performance. This is often quite high, with connections between servers and switches operating between 10 Gbps to 100 Gbps.

The external bandwidth of a data center, which is the total capacity for data transfer between the data center and the outside world, can also vary widely. Smaller data centers might have external bandwidth in the range of 1 to 10 Gbps, while large data centers might have external bandwidth in the order of Tbps.

From the perspective of the user, the most important measure of performance is the response time, the time it takes to receive a response. The response time factors in travel time through the internet to the data center, processing time at the data center, travel time back through the internet and any delays or wait time that occurs in between.

RESPONSE TIME IS THE TIME IT TAKES FOR A USER TO RECEIVE A RESPONSE FROM A SERVER TO A REQUEST.

DEPENDABILITY

Regardless of response time, throughput and bandwidth, users depend on data centers to run without any faults or failures. Users do not appreciate errors nor the failure of a server to respond. The dependability of a system such as a data center, refers to the extent to which it can operate without fault or failure.

DEPENDABILITY IS THE EXTENT TO WHICH A SYSTEM WILL OPERATE WITHOUT FAULT OR FAILURE.

Common reasons for faults or failures include over-heating, power outages, security breaches and disasters such as fire.

- 1. Over-heating:** Servers generate lots of heat due to the high volume of computational work they have to perform. Therefore, cooling systems such as Computer Room Air Conditioning (CRAC) units, Computer Room Air Handler (CRAH) units, and liquid cooling systems are essential to maintaining the proper operating temperature and preventing hardware failure.
- 2. Power outage:** Power disruptions can cause significant issues for data centers, including hardware damage and loss of data. Uninterrupted Power Supply (UPS) systems and backup generators are used to maintain power to the data center during outages.
- 3. Security Breach:** Physical and cyber security breaches can lead to data loss, service disruption, and significant harm to the data center's reputation. Physical and cyber security measures are implemented to protect the data center.
- 4. Disasters:** Natural disasters and accidents like fires or floods can cause catastrophic damage to data centers. Fire suppression systems, flood defenses, and disaster recovery protocols including off-site data backups and redundant systems are crucial for protecting the data center and ensuring uninterrupted service.

THE INTERNET OF THINGS

While the cloud has taken processing and storage capabilities beyond the confines of personal computers, the internet of things (IoT) extends input and output functionalities to a wide array of devices beyond traditional computers.

Computers are not the only devices connected to a network. A network may also connect other things. The term “things” refers to devices that can link to a network and either provide input or generate output forming an internet of things. IoT is a system of interrelated devices that are connected to a network and can send or receive data without the need for human interaction. These devices can include anything from small sensors to large appliances.

THE **IoT IS A SYSTEM OF INTERRELATED DEVICES CONNECTED TO A NETWORK AND TRANSMIT DATA.**

In the context of the IoT, a personal area network, PAN, can be created to connect a variety of smart devices to a central hub, often a smartphone or a personal computer.

EXAMPLE 3.3

Sam lives in a smart home. Using a central hub, a small computer designed for the purpose of smart home management, Sam can control and automate his home particularly because his home is full of smart things connected by a network to the hub.

Here are some of the smart things Sam has: smart watch, smart coffee machine, smart refrigerator, smart air conditioner, smart curtains, smart cameras and smart lights.

At 6:30 am, Sam’s smartphone alarm, synced with his sleep cycle via his smartwatch, wakes him up at the lightest point in his sleep, making waking

up less of a struggle. The moment he switches off the alarm, his smart coffee machine in the kitchen starts brewing his morning coffee.

After getting ready, Sam walks into the kitchen, where his smart refrigerator has already identified that he's running low on milk and eggs. The fridge automatically adds these items to Sam's grocery list on his smartphone, and it even suggests a healthy breakfast recipe based on the items currently available.

As Sam leaves his house for work, he doesn't need to worry about whether he has switched off all the lights or the AC. His smart home hub, which is connected to all the IoT devices in the house, knows when he's leaving due to his smartphone's GPS. It automatically turns off unnecessary lights and adjusts the thermostat to an energy-saving mode.

During the day, Sam's home security system keeps an eye on the house. The smart cameras and sensors monitor for any unexpected activity. If the cameras detect movement, Sam instantly receives a notification on his phone. He can watch a live feed and, if necessary, sound an alarm or contact the authorities, all from his phone.

Once Sam heads back home, his smart home hub detects his smartphone's location and begins to prepare for his arrival. The thermostat adjusts to a comfortable temperature, and the lights turn on as he walks in. As he steps into the house, his smart home hub greets him and briefs him about his schedule for the next day, the weather forecast, and any unread emails or messages.

At night, Sam can use his voice to control his home. A simple command like, "Good night, Home," can cause the lights to dim, the doors to lock, the alarm system to activate, and his favorite relaxing playlist to start playing softly in the background.

In this manner, each device in Sam's smart home plays a role in automating routine tasks, enhancing his comfort, saving energy, and providing him with peace of mind about his home's security. All these devices form a network around Sam, working together to make his life easier and more efficient.

An IoT is not restricted by distance, as long as there is network coverage. Various sensors and smart devices can connect to a central computer. Areas as wide as farms and cities can be covered by IoT. Here are two examples.

THE **IoT** IS NOT RESTRICTED BY DISTANCE AS LONG AS THERE IS NETWORK COVERAGE.

EXAMPLE 3.4

Meet Farmer Maya, an innovative agriculturist using smart farming to optimize her operations. At dawn, Maya's weather station, equipped with sensors, provides a precise forecast, helping her plan the day's tasks. Her smart irrigation system, connected to soil moisture sensors, already knows which parts of her farm need water. These sensors feed data to the system, which autonomously controls the water flow, ensuring each crop gets the right amount of hydration while conserving water.

Maya has drones equipped with infrared cameras. When flown over her fields, they provide valuable data about crop health, identifying potential disease outbreaks or pest infestations before they become visible to the human eye.

Livestock wear connected devices that monitor their health and location. If a cow's activity level drops or it strays from the herd, Maya gets a notification on her phone.

At the end of the day, all data collected is sent to Maya's computer. Thanks to her smart farm, Maya ensures that her crops grow well, her livestock are healthy and that she uses her water resources efficiently.

EXAMPLE 3.5

Director Dave, is in charge of a city's waste management department and has implemented smart technology to optimize garbage collection.

To start the day, Dave receives an automated report generated by smart sensors installed in public garbage bins throughout the city. These sensors monitor the fill levels of the bins and report back when they're nearing capacity. This real-time data helps Dave's team plan their collection routes, ensuring they only visit bins that need to be emptied and reducing unnecessary trips.

Dave's garbage trucks are equipped with GPS and route optimization software. This allows the central system to direct each truck along the most efficient route, saving time and reducing fuel consumption.

The smart waste bins also have sensors that identify different types of waste for recycling. This helps the city to improve its recycling rate and reduces the volume of waste sent to landfills.

With this smart waste management system, Dave has made garbage collection more efficient, improved recycling rates, and made his city cleaner and more sustainable.

Career Tips

IoT Solution Architects: These individuals design comprehensive IoT solutions, ensuring various components like devices, cloud services, and connectivity solutions work seamlessly together.

IoT Hardware Engineers: They are responsible for designing and developing the hardware aspects of IoT devices, such as sensors, circuits, and embedded systems.

SUMMARY

The internet brought about the cloud and the internet of things.

The cloud is a system of servers that can be accessed over the internet. The cloud offers additional storage and processing services.

Additional storage can be obtained by using memory cards, external drives and cloud-based storage services. The advantages of cloud-based storage are its accessibility from any device with internet access and automatic backups ensure the most recent information is kept secure. The disadvantages are its dependence on the internet and service providers for accessibility and security, as well as its potential recurring cost.

A data center is a facility that houses servers working as an integrated system serving the requests of many users. It is a system of integrated components, such as servers, switches, power units, cooling systems and security systems.

The server is the heart of a data center. It is a powerful computer designed to handle large amounts of data and run applications that serve other computers and their users across a network.

Throughput is a measure of server performance and is defined as the number of requests processed per second. Throughput is dependent on bandwidth which is the amount of data that can be sent or received. Data centers have both an internal and external bandwidths. Servers communicate with one another inside the data center and they also communicate with clients via an external network.

From the perspective of the user, the most important measure of performance is the response time, the time it takes to receive a response.

The performance of a server is enhanced by parallel processing. Servers are arranged in clusters, which allows the for parallel processing. Servers in a cluster work simultaneously to perform multiple tasks for multiple users.

Dependability is the extent to which a system will operate without fault or failure, which can be caused by over-heating, power outages, security breaches and disasters such as fire.

The IoT is a system of interrelated devices connected to a network and transmit data autonomously. The IoT extends input and output functionalities to a wide array of devices beyond traditional computers. All these devices are connected to a central hub.

The IoT made smart homes, smart farms and smart cities possible.

Activity 3.1

Read the following statements, and write on the line provided for each statement 'T' if it is true and 'F' if it is false.

T/F	STATEMENT
<u>F</u>	The cloud does not depend on the internet.
<u>T</u>	Both storage and processing can be provided as cloud-based services.
<u>T</u>	A data center is a facility with an integrated system of servers.
<u>T</u>	Servers, like other computers, have the four basic components of input, output, memory and processing.
<u>F</u>	Throughput is a measure of dependability.
<u>T</u>	Bandwidth refers to the amount of data transmitted per second.
<u>T</u>	To improve performance servers are arranged in clusters that perform parallel processing.
<u>T</u>	Dependability is the extent to which a system will operate without fault or failure.
<u>T</u>	The IoT is a system of interrelated devices that are connected and can transmit data.
<u>F</u>	The IoT only includes devices in close proximity.

Activity 3.2

Answer the questions below.

1. What is the “cloud” in cloud computing?
 - a) A weather system
 - b) A system of servers accessed over the Internet
 - c) A data storage device

2. Which of the following is not an advantage of cloud storage?
 - a) Accessible from any device with internet
 - b) Automatic backup and syncing
 - c) Doesn't require an internet connection

3. Which of the following tasks can be performed by a cloud-based service?
 - a) Only storage
 - b) Only processing
 - c) Both storage and processing

4. Which additional storage option provides the greatest space?
 - a) cloud-based storage
 - b) memory card
 - c) external drive

5. Which additional storage option depends on the internet?
 - a) cloud-based storage
 - b) memory card
 - c) external drive

Activity 3.3

Answer the questions below.

1. What is the main component of a data center?
 - a) Cooling system
 - b) Network switch
 - c) **Server**

2. What does the term “throughput” refer to in the context of a server?
 - a) The amount of data a server can store
 - b) **The number of tasks a server can complete in 1 second**
 - c) The speed at which a server can download data

3. What is “bandwidth” in the context of data center performance?
 - a) The physical width of the cables connecting servers
 - b) The number of servers in a data center
 - c) **The amount of data that can be sent or received per second**

4. What is the measure of performance from the user’s perspective?
 - a) Throughput
 - b) Bandwidth
 - c) **Response time**

5. What does “dependability” in the context of a data center refer to?
 - a) The amount of data a server can store
 - b) **The extent to which a system will operate without fault or failure**
 - c) The speed at which a server can download data

Activity 3.4

Answer the questions below.

1. What does the term “Internet of Things” refer to?
 - a) The network connecting all the computers in the world
 - b) A network of interrelated devices that can transmit data without the need for human interaction
 - c) A network of satellites providing global internet coverage
2. What can smart home devices do in the context of IoT?
 - a) Automate routine tasks
 - b) Enhance comfort
 - c) Both a and b
3. What is a central hub in the context of IoT?
 - a) A device that connects to other devices on a network and manages them
 - b) The main server in a data center
 - c) The main router in a network
4. How does smart farming help in conserving water?
 - a) By watering all crops equally irrespective of their needs
 - b) By using soil moisture sensors to autonomously control the water flow
 - c) By only watering the crops during the night
5. How can a city’s waste management department benefit from IoT?
 - a) By having to manually check the fill levels of the bins
 - b) By planning their collection routes based on real-time data about the fill levels of the bins
 - c) By increasing the volume of waste sent to landfills

Activity 3.5

Answer the questions below.

1. How has the internet contributed to the evolution of the basic model for a computer?

The internet has allowed for the decoupling and geographical dispersal of the four basic computer functions: input, output, memory, and processing. This has given rise to new computing models such as cloud computing and the Internet of Things.

2. What are some potential advantages and disadvantages of using a cloud-based storage service?

Advantages of cloud-based storage services include accessibility from any device with internet, automatic backup and syncing of data. Disadvantages include the requirement of an internet connection to access data, potential security and privacy concerns, and the recurring cost for extra storage.

3. How can cloud-based services enhance the functionality of your computer?

Cloud-based services can provide additional storage and processing power beyond the capabilities of one's personal computer. This allows for more complex tasks to be performed and larger amounts of data to be stored than what could be achieved with the computer alone.

Activity 3.6

Answer the questions below.

1. What are the main components of a data center?

The main components of a data center are servers, switches, power units, cooling systems and security systems.

2. How can bandwidth limitations impact the performance of a data center?

Bandwidth limitations can limit the amount of data that can be sent to and processed by the servers, reducing their throughput. If the bandwidth is insufficient, data delivery to the servers would be too slow, and the servers won't be able to process tasks at their full capacity.

3. What measures can be taken to ensure the dependability of a data center?

Measures to ensure the dependability of a data center include cooling systems to prevent overheating, Uninterrupted Power Supply systems and backup generators to maintain power during outages, physical and cyber security measures to protect against security breaches, and fire suppression systems,

Activity 3.7

Answer the questions below.

1. How do smart home devices improve a user's quality of life?

Smart home devices improve a user's quality of life by automating routine tasks, enhancing comfort, providing security, and increasing efficiency. They can adjust to the user's habits and preferences, such as brewing coffee when the alarm goes off, or suggesting recipes based on available ingredients.

2. What is one way smart farming benefits the environment?

Smart farming benefits the environment by optimizing water usage, with soil moisture sensors controlling the water flow to ensure each crop gets the right amount of hydration. This helps in conserving water.

3. How does a smart waste management system contribute to a cleaner and more sustainable city?

A smart waste management system contributes to a cleaner and more sustainable city by optimizing garbage collection, reducing unnecessary trips, and saving time and fuel. Sensors in the bins monitor fill levels and enable collection only when necessary. They also identify different types of waste for recycling, improving recycling rates and reducing the volume of waste sent to landfills.



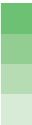
Activity 3.8

Cloud Storage Activity: Sign up for a free cloud storage service such as Google Drive or Dropbox. Upload a document or picture from your computer, and then access and download it using a different device. This will demonstrate the concepts of cloud storage and accessibility from multiple devices.



Activity 3.9

Exploring Data Center Virtual Tour: Many data centers offer virtual tours online (like Google data centers). Explore these tours to understand the scale, organization, and components of a data center, including servers, cooling systems, and security measures. Then discuss or write a report on your findings.



Activity 3.10

IoT Smart Home Design: Divide students into groups and ask each group to design a smart home using IoT devices. Students can research different types of IoT devices, decide what their smart home would include, and explain how the devices would communicate with each other and the central hub. This project can be presented as a diagram, a report, or a presentation.



Activity 3.11

Creating a Smart City Map: Divide students into groups and ask each group to design a “smart city” that utilizes IoT technology. Students can research different applications of IoT in urban environments (like smart traffic lights, waste management systems, air quality sensors) and incorporate these into their city. They can present their city map and explain how each IoT element contributes to the efficiency, sustainability, or livability of the city.

LESSON 4: INFORMATION SYSTEMS

In previous lessons we have explored how humans interact with their computers and how these computers communicate with each other across networks. We have explored how functions such as input, output, storage and processing can occur over networks as in the case of the cloud and the IoT. In this lesson, we will explore how all these can be brought together, humans, computers, networks, cloud and IoT, to form larger systems to handle information.

FUNCTIONS OF INFORMATION SYSTEMS

Every information system serves a particular purpose, such as managing a school or providing navigation services. Regardless of the purpose, all information systems collect data, store data, and process data in order to extract meaningful information and share it to help guide decision-making.

**AN INFORMATION SYSTEM PERFORMS FOUR PRIMARY FUNCTIONS:
DATA COLLECTION, STORAGE, PROCESSING, AND SHARING.**

An information system performs four primary functions: data collection, data storage, data processing, and information sharing.

Collection: An information system gathers data from various sources. This could be through direct user input or automated data collection from sensors as in the case of the IoT.

Storage: The collected data is stored mainly in the form of databases on servers making the data accessible over a network. Storage may be on a local server or it may be on a server based in the cloud.

Processing: This is where the data is transformed into useful information. The system might organize the data, perform calculations, apply various algorithms, or create visualizations. Depending on where the data is stored, processing may occur at the level of a local server or in the cloud. The aim is to turn raw data into meaningful information that supports decision-making or other activities.

Sharing: Information systems are designed to distribute and share data among users. This sharing ensures that the necessary data is available when and where it's needed. Information systems have multiple users with whom information is shared. Access to the information may be restricted to ensure the security and confidentiality of data and information.

To better understand information systems let's explore two types: a management information system and a geographic information system.

SCHOOL INFORMATION SYSTEMS

A school information system is a system designed to collect, store, process and share information pertinent to all stakeholders in a school, from students and their parents to employees and their administrators.

EXAMPLE 4.1

We will explore a SIS from the perspectives of a student, parent, teacher, principal and the SIS administrator.

Story 1: Student - Sara

Sara is a student in seventh grade. Every morning, she checks her SIS mobile app to see her schedule for the day. After school, she checks for assignments her teachers have posted. Once she completes her homework, she uses the SIS to submit her assignments. When she finishes a big project or a test, she eagerly awaits the grade on the SIS - it's the first place her grades are posted.

In addition, Sara has a smart ID card. She uses it to swipe in and out of school

that way the SIS automatically takes attendance. She also uses it to purchase food and drinks from the school's cafeteria.

Story 2: Parent - Ahmed

Every morning, Ahmed checks his children's timely arrival to school using his SIS mobile app. At work, he frequently checks their schedules and academic progress using his desktop. One day, he noticed a dip in his son's math scores. He messaged the teacher via SIS, and arranged a virtual meeting to discuss improvement strategies. School announcements are easily accessed on SIS, keeping Ahmed informed. Despite his hectic schedule, SIS enables Ahmed to stay closely connected with his children's education.

Story 3: Teacher - Mr. Johnson

Mr. Johnson starts his day by checking his class rosters on the SIS, noting any changes. Throughout the day, he uses the SIS to take attendance, input grades, and assign homework. He appreciates how he can quickly message students or parents through the SIS to discuss grades or concerns. At the end of the day, Mr. Johnson uses information from the SIS to monitor students' progress and customize his lesson plans accordingly.

Story 4: Principal - Mrs. Gomez

Mrs. Gomez begins her day by reviewing reports from the SIS, looking at attendance records, academic performance, and other key metrics. The SIS helps her identify patterns, such as if a particular class is struggling with attendance or a specific subject. She also uses the SIS to communicate with the school community, whether it's a reminder about an upcoming event or a critical update.

Story 5: System Administrator - Mr. Lee

Mr. Lee oversees the SIS, ensuring it runs smoothly. His morning starts with system checks and addressing any issues. Throughout the day, he manages user accounts, granting necessary access to new staff, or students. Mr. Lee also installs updates and troubleshoot issues. He plays a crucial role in training staff on how to use the SIS effectively.

GEOGRAPHIC INFORMATION SYSTEMS

A Geographic Information System (GIS) is a type of information system that is specifically designed to collect, store, process, and share geographical data. GIS allows users to visualize data primarily in the form of maps. It also provides users with information about various locations on the map, such as names of streets, traffic congestion, photographs, and directions to go from one place to another.

EXAMPLE 4.2

Google Maps is a commonly used geographic information system. We will explore it from the perspectives of a user, data analyst, street view driver, 3D modeling specialist and software engineer.

Story 1: User - Sofia

Sofia uses Google Maps in her everyday life. She starts her day by checking the fastest route to her office, taking into account real-time traffic updates. During lunch, she uses Google Maps to find a new restaurant nearby. After work, she uses Street View to virtually explore a neighborhood where she's considering renting a new apartment. For Sofia, Google Maps and its GIS underpinnings make her daily life more convenient and informed.

Story 2: Data Analyst - Kelly

Kelly starts her day by pulling new satellite images from multiple sources. She

uses GIS software to correct and enhance these images for clarity. By overlaying new images on old ones, she detects significant changes, like deforestation or urban growth. She updates Google Maps with the new data, helping to ensure that users have the latest information when planning routes or exploring areas.

Story 3: Street View Driver - Mario

Mario is responsible for capturing street-level images for Google Maps. His van, equipped with a multi-lens camera, captures 360-degree views of the streets he drives. He spends his day driving assigned routes, often in new cities or areas where significant changes have occurred. The images he captures are sent to Google for processing and integration into the Street View feature.

Story 4: 3D Modeling Specialist - Mei

Mei creates realistic 3D models of buildings and landmarks for Google Maps. She uses high-resolution aerial images and elevation data as her primary sources. Through GIS software, she's able to translate this data into detailed 3D models. When a user explores a city using the 3D feature in Google Maps, they're seeing Mei's work.

Story 5: Software Engineer - Lucas

Lucas works on the backend of Google Maps, making sure everything runs smoothly. His day often involves optimizing algorithms that calculate routes, improving the map rendering speed, or ensuring that the huge amount of data Google Maps uses is stored efficiently. Lucas' work doesn't always have a visible output on the Google Maps interface, but without it, the service wouldn't be nearly as fast or reliable.

COMPONENTS OF INFORMATION SYSTEMS

Every information system despite its complexity is composed of four components: hardware, software, data and people. These four components are used to perform the functions of data collection, storage, processing and sharing.

**AN INFORMATION SYSTEM HAS FOUR PRIMARY COMPONENTS:
HARDWARE, SOFTWARE, DATA, AND PEOPLE.**

Hardware: The physical components of an IS, including computers, network equipment, and IoT devices. Computers include all types personal computers as well as servers, whether they are local servers or part of a data center. Network equipment includes switches and routers. While IoT devices can be a very wide variety of devices that range from simple sensors to smart cars.

The school information system hardware includes all the personal computers of its users as well as the school server and the smart ID carried by students. The geographic information system includes a tremendous number of hardware that range from satellites to every smartphone running the Google Maps app. Also included in the Google Maps information system are the data centers that store and process its data, and share its information with all its users.

Software: The programs and applications used to process and manage information. These also include the databases that contain all the data. Some information systems consist of one application that performs all the functions, whereas others may consist of multiple applications each performing a particular task and collectively forming an information system.

In the case of the school information system, it consists of one application which all users access. Whereas the geographic information system consists

of multiple applications, Google Maps accessible by the general public, as well as data analysis software, photo-editing software and 3D modeling software to name a few.

Data: The raw, unprocessed facts and figures without any added interpretation or analysis. Data is simply the input collected either from users entering it using the application or automatically collected by IoT devices.

School Information Systems (SIS) Data

Student Data: This includes data such as student names, IDs, grades, test scores, attendance, demographic data, and any other information related to individual students.

Class Data: Information about each class, including the teacher, room number, subject, schedule, and enrolled students.

Teacher Data: Details about teachers, such as their name, contact information, subjects taught, schedules, and any other relevant information.

Parent Data: Parent or guardian names, contact information, and their relation to students.

School Calendar: Important dates for the academic year, including holidays, start and end dates of terms, and dates of significant events.

Administrative Data: Information about school operations, including budgets, resource allocation, staff details, and other operational data.

Geographic Information Systems (GIS) Data

Satellite Imagery: Images captured by satellites that provide a top-down view of the Earth's surface.

Aerial Photographs: These are photographs taken from aircraft and drones,

which can often provide more detail than satellite imagery.

Maps: Both digital and paper maps can serve as data for a GIS.

GPS Data: Data from Global Positioning System devices can provide very accurate spatial data.

Street View Data: Images and data collected by vehicles equipped with 360° cameras, like those used for Google's Street View feature.

Environmental Sensor Data: Data from sensors that measure environmental conditions like temperature, humidity, soil moisture, etc., often used in environmental or agricultural applications of GIS.

People: Include both the end users who interact with the information system and the people that perform the tasks that keep the system operational, such as technicians, system administrators and software engineers to name a few.

The school information system includes its users, the students, parents, teachers and principle, as well as system administrators and technicians that help keep the system operational. The geographic information system has a far greater number of users, namely any person who has the app, and many other people that work to keep the system running and updated, such as street car drivers, data analysts, 3D modeling specialists and data engineers.

Information systems play an important role in our personal and professional life and demonstrate the importance of information in the decision-making processes of our lives.

SUMMARY

Information systems perform four primary functions: data collection, storage, processing and sharing of data and information.

The overall goal of information systems is to guide decision-making by collecting data, processing it to extract meaningful information, and to share both data and information with users in such a way as to assist them with making decisions.

Information systems have four primary components: hardware, software, data and people. The hardware includes all IoT devices, personal computers, servers and network hardware such as switches and routers. The software includes the programs and applications that are used to collect, store, process and share data and information. Data is a critical component of an information system and is collected in a database as input from IoT devices and users. It is processed to extract useful information. People are part of the information system. They act as users and sources of data input as well as professionals that ensure the information system operates properly.

There are many types of information systems and each has a purpose. A school information system collects, stores, processes and shares data and information with members of the school community to assist each in making decisions related to education and the educational process. A geographic information system collects, stores, processes and shares data and information with the public to assist people in making decisions related to places and travel.

Career Tips

Software Engineers: These individuals design and build information systems. They are responsible for creating the software components that make up the system.

Information System Managers: They are responsible for overseeing the implementation of new systems, managing IT teams, and allocating resources.

Activity 4.1

Read the following statements, and write on the line provided for each statement 'T' if it is true and 'F' if it is false.

T/F	STATEMENT
<u>T</u>	An information system is responsible for collecting data.
<u>T</u>	Data processing is used to extract meaningful information.
<u>F</u>	An information system does not share information with its users.
<u>T</u>	An information system may incorporate IoT devices.
<u>F</u>	Cloud storage is not an option for information systems.
<u>T</u>	Any device that collects data can be used as component of an information system.
<u>F</u>	An information system cannot include more than one software.
<u>T</u>	A database must remain secure.
<u>F</u>	People do not play an important role in an information system.
<u>T</u>	Information systems presents information to users to guide their decisions.

Activity 4.2

Answer the questions below.

1. What are the four primary functions of an information system?
 - a) Data collection, data mining, data processing, information sharing
 - b) **Data collection, data storage, data processing, information sharing**
 - c) Data collection, data storage, data analysis, data sharing

2. What is the primary purpose of an information system?
 - a) **To guide decision-making**
 - b) To store data
 - c) To collect data

3. Which of the following is a source of data for a school information system?
 - a) teacher
 - b) smart ID
 - c) **Both a and b**

4. Which of the following is a source of data for a geographic information system?
 - a) satellite
 - b) map
 - c) **Both a and b**

5. What is the primary way information systems help with decision-making?
 - a) By collecting data from users
 - b) **By processing raw data into meaningful information**
 - c) By storing data in databases

Activity 4.3

Answer the questions below.

1. What is the role of a system administrator in a school information system?
 - a) To enter data
 - b) To oversee the system and ensure it runs smoothly
 - c) To analyze data

2. What is the role of a data analyst in a Geographic Information System like Google Maps?
 - a) To drive the Street View vehicle
 - b) To pull new satellite images and enhance them for clarity
 - c) To create 3D models of buildings

3. What are the components of an information system?
 - a) Hardware, software, information
 - b) Software, data, IoT devices, cloud
 - c) Hardware, software, data, people

4. What kind of devices does the 'hardware' component of an information system include?
 - a) servers, personal computers
 - b) IoT devices, routers
 - c) Both a and b

5. What do Geographic Information Systems (GIS) primarily visualize data in the form of?
 - a) Charts
 - b) Maps
 - c) Tables

Activity 4.4

Answer the questions below.

1. What are the main functions of an information system?

The main functions of an information system are to collect data, store data, process data into meaningful information, and share information.

2. What are some types of data stored in a school information system?

A school information system stores various types of data including student data (names, IDs, grades, attendance), class data (teacher, room number, subject, schedule), teacher data, parent data, school calendar, and administrative data (budgets, resource allocation).

3. What are some types of data for a geographic information system?

Satellite images, aerial photographs, maps, GPS data, Street view data, environmental sensor data

Activity 4.5

Answer the questions below.

1. What are the main components of an information system?

The main functions of an information system are to hardware, software,
data, and people.

2. What is the role of IoT devices in information systems?

IoT devices play a crucial role in information systems by automating data
collection. These devices can collect data from their environment and send it
to the system for processing and analysis.

3. What role do people play in an information system?

People are both the end users who interact with the system and the
professionals who keep the system operational. They contribute to data input
and use the output information for various purposes. Professionals such as
system administrators and software engineers ensure smooth operation and
maintenance of the system.

Activity 4.6

Interview an IT Professional: Students can prepare and conduct an interview with a school's IT administrator or a GIS specialist. They can ask about the everyday challenges, maintenance requirements, and the skills needed to operate and manage these systems.

Activity 4.7

Build a Mock School Information System: Students can design and create a basic mock-up of a school information system. They can define what data should be collected (such as student names, grades, attendance), decide how to store it, define methods for processing the data, and consider how it should be shared with users. This activity will help them understand the practical aspects of an information system.

Activity 4.8

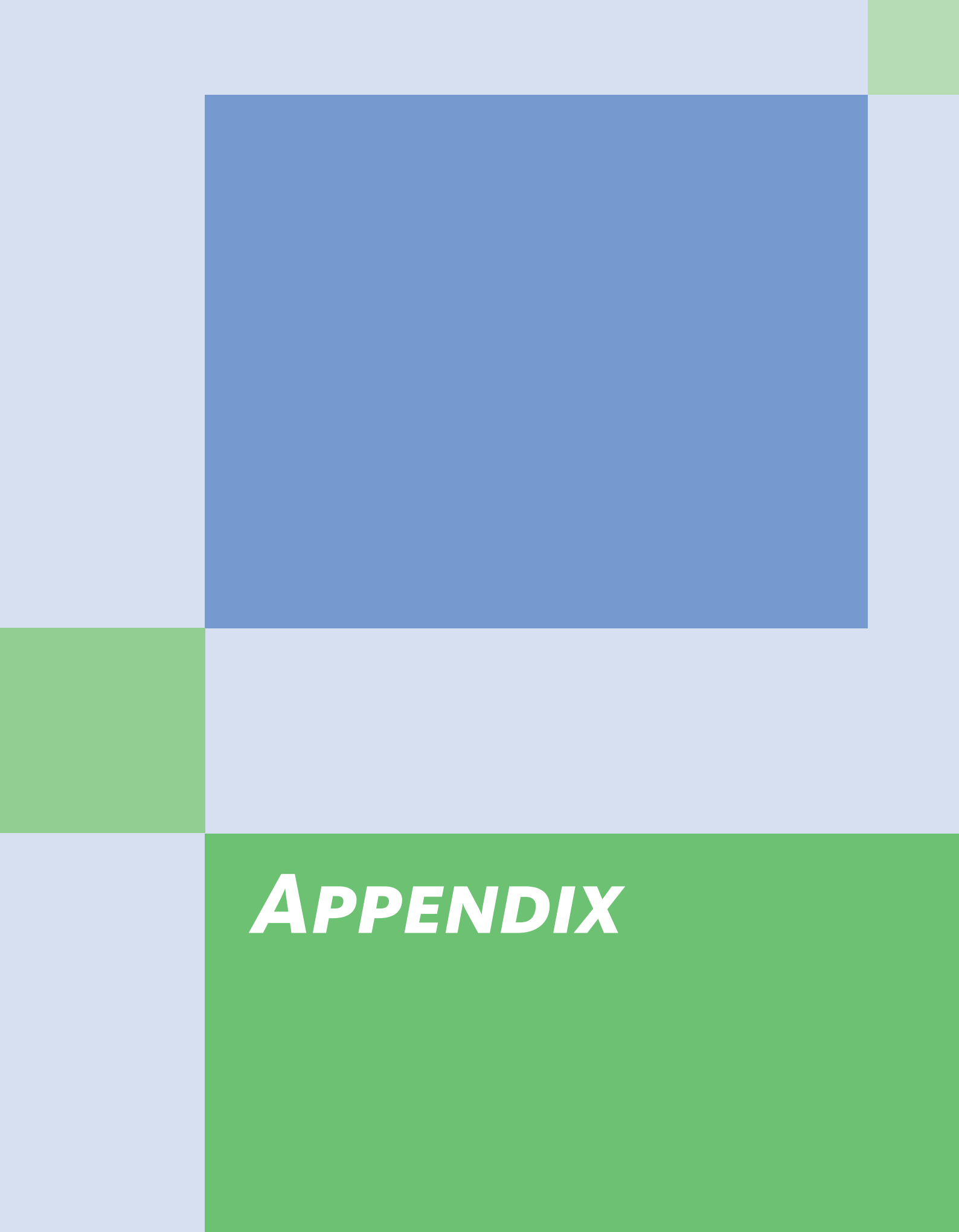
Explore Google Maps: Students can explore the various features of Google Maps (such as Street View, satellite imagery, traffic updates, etc.) and report on how these features help in their daily life. This activity will help them understand how a Geographic Information System works.

Activity 4.9

Discussion on Privacy and Security: Organize a discussion or a debate on the privacy and security concerns related to information systems. It will make students think critically about the ethical aspects of data collection, storage, and sharing.

Activity 4.10

Design User Interface: Students can design the user interface for a new feature in the school information system or Google Maps. They can then present their design to the class, explaining how users would interact with the system via this interface. This activity encourages students to think about the user experience aspect of information systems.

The background of the page is a light blue grid. A large blue rectangle is positioned in the upper left, and a green rectangle is in the lower right. The word 'APPENDIX' is written in white, bold, italicized capital letters on the green rectangle.

APPENDIX

GLOSSARY

TERM	DEFINITION	PAGE
Accelerometer	A sensor in devices like smartphones that detects straight-line motion, such as moving up and down, forward and backward, or side to side.	p. 58
Acknowledgement	A signal sent back to confirm that a message was received successfully.	p. 151
Actuator	An output device that performs an action by converting signals into movement, like a motor in a printer or robot.	p. 59
Action Potential	The internal electrical signal used by the human nervous system to transmit information between body parts.	p. 55
Adder	A logic circuit that performs addition on two input signals and outputs their sum.	p. 78
Address	A specific location in memory or registers where data is stored or retrieved.	p. 82
Address Bus	A set of lines in the bus system that carry the addresses of where data should be read from or written to.	p. 103
Administrative Data	Information about how a school or organization runs, such as budgets, staff details, or resources.	p. 198
Aerial Photographs	Pictures taken from aircraft or drones that provide detailed views of the Earth's surface, often used in GIS.	p. 199
Algorithm	A set of steps or rules a computer system follows to process data and solve a problem.	p. 193
Application (App)	A stored program that performs a specific task, such as writing an essay, playing music, or browsing the web.	p. 20–21
Architecture	The basic structure of a computer system, including the processor, memory, input/output devices, a bus, and a clock.	p. 102
Arithmetic Logic Unit (ALU)	The part of the processor that performs calculations and logical operations on data.	p. 81
Autonomous Robot	A robot that can make decisions and act on its own without human control.	p. 128
Bandwidth	The amount of data that can be sent or received in one second across a network, usually measured in bits per second (bps).	p. 175
Binary	A number system using only two values, 0 and 1, which computers use to represent information.	p. 77
Bit	The smallest unit of information in computing, representing either a 0 or a 1.	p. 77

Bluetooth Adapter	A type of adapter that uses short-range radio waves to connect nearby devices, such as headphones, speakers, or keyboards.	p. 146
Bus	A shared communication pathway that connects all components of a computer so they can send and receive signals.	p. 103
Byte	A group of 8 bits that together can represent more complex information.	p. 77
Cache	A small, high-speed memory placed close to the processor to store frequently used data and instructions for quick access.	p. 109
Caesar Cipher	An old encryption method where each letter in a message is shifted by a fixed number of positions in the alphabet.	p. 154
Capability	How powerful a computer is, based on the complexity of tasks it can handle and the speed of its calculations.	p. 34
Checksum	A number calculated from data before transmission, used to check if the data arrived correctly without errors.	p. 152
Class Data	Information about a class, including the teacher, subject, schedule, and enrolled students.	p. 198
Clock	A part of the computer that sends out regular timing signals to keep all components working in sync.	p. 104
Clock Cycle	One complete vibration of the clock's quartz crystal, which acts like a "go" signal for moving data to the next step.	p. 104
Clock Frequency	The number of clock cycles per second; higher frequency usually means faster performance.	p. 106
Cluster	A group of interconnected servers that work together at the same time to handle many user requests in parallel.	p. 175
Command-Line Interface (CLI)	A type of user interface where the user types text commands to make the computer perform tasks.	p. 61
Comparator	A logic circuit that compares two inputs and outputs whether one is greater than, less than, or equal to the other.	p. 80
Communication	The exchange of information between two or more people, often using methods like messaging or email.	p. 18
Computer	A technology designed to process information according to a stored program.	p. 10, 15
Computational Power	The ability of a computer to process large amounts of information and perform calculations very quickly.	p. 34
Concurrent Processing	When a computer quickly switches between tasks to give the impression that multiple tasks are running at the same time.	p. 175
Control Bus	The part of the bus that carries control signals, including timing signals from the clock.	p. 103
Control Unit (CU)	The part of the processor that directs signals, decodes instructions, and activates the correct logic circuits.	p. 84

Core	An individual processing unit inside a CPU or GPU; modern processors may have multiple cores working in parallel.	p. 107
Data	Raw facts or numbers that can be processed to create meaningful information.	p. 15
Data Bus	The part of the bus that carries the actual data being transferred between components.	p. 103
Data Center	A large facility that houses many servers and related systems (cooling, power, security) to provide cloud-based services.	p. 173
Database	An organized collection of data stored on servers that can be accessed and managed by software.	p. 192
Dependability	How well a system, like a data center, can keep running without faults or failures.	p. 177
Destination Address	The location where the result of an operation is stored after processing.	p. 83
Digital Electronic Signal	The electrical signals used inside computers, which can only be high (1) or low (0).	p. 77
Directory (Folder)	A virtual container used by the computer's filing system to organize files; it can also contain other directories (subdirectories).	p. 73
Encryption	The process of changing a readable message into a secret code so that only authorized people can read it.	p. 153
Environmental Sensor Data	Measurements collected by sensors about conditions like temperature, humidity, or soil moisture.	p. 199
Error Detection	Techniques, like checksums and acknowledgements, used to check if data was transmitted correctly.	p. 151–152
Ethernet NIC	A network interface card that connects computers using cables to exchange data.	p. 146
External Bandwidth	The total data transfer capacity between a data center and the outside world (users, internet).	p. 176
Field	A part of a machine-language instruction that carries specific information, such as the opcode or an address.	p. 87
File	A unit of stored information, such as a document, photo, or program, that has a name and address.	p. 19
Filename	The name given to a file so that it can be identified.	p. 19
Frequency-Hopping Spread Spectrum (FHSS)	A method used by Bluetooth to send signals by rapidly switching between radio frequencies to reduce interference.	p. 146
General-Purpose Computer	A computer designed to perform many different tasks using different applications, like desktops, laptops, tablets, and smartphones.	p. 32–33
Geographic Information System (GIS)	A type of information system designed to collect, store, process, and share geographic data, often shown as maps.	p. 195

Google Maps	A widely used GIS application that provides maps, directions, traffic updates, and features like Street View.	p. 194–195
GPU (Graphics Processing Unit)	A processor designed specifically to handle graphics-related instructions, often working alongside the CPU.	p. 107
Graphical User Interface (GUI)	A user interface that uses windows, icons, and menus so people can interact with computers more visually and easily.	p. 62
Gyroscope	A sensor that detects rotation, like rolling, spinning, or tilting movements of a smartphone.	p. 58
Hardware (in IS)	The physical parts of an information system, such as computers, servers, network devices, and IoT equipment.	p. 197
Hop Count	The number of routers a signal must pass through to reach its destination.	p. 147
Human-Computer Interaction (HCI)	A field of study that focuses on how people design, use, and interact with computers and technology.	p. 122
Human-Computer Interface	The boundary where humans and computers exchange signals, usually through input and output devices.	p. 122
Human-Machine Interface (HMI)	The point where humans interact with machines that have embedded computers, like cash registers or smart devices.	p. 126
Information	Any meaningful content that tells us something about ourselves or the world, such as words, images, or sounds.	p. 12
Information System (IS)	A system that collects, stores, processes, and shares data to provide useful information and support decision-making.	p. 192
Input Device (Sensor)	A device that converts signals from the outside world, like light, sound, or pressure, into internal digital signals a computer can use.	p. 57
Input/Output (I/O) Loop	The continuous cycle where a computer's output becomes the human's input, and the human's output becomes the computer's input.	p. 123
Instruction	A command given to the processor that tells it what operation to perform and on which data.	p. 81
Instruction Cache (I-Cache)	A cache memory that stores instructions separately, making it faster for the processor to find the next instruction.	p. 110
Instruction Cycle	The repeated process of fetching, decoding, and executing instructions in a processor.	p. 91
Instruction Set	The complete collection of instructions that a processor is designed to understand and carry out.	p. 83
Interface	The boundary where a system communicates with the outside world by exchanging signals.	p. 56
Internet of Things (IoT)	A system of devices (“things”) connected to a network that can send or receive data automatically without human input.	p. 178
Internal Bandwidth	The data transfer capacity between servers inside a data center.	p. 176

LAN (Local Area Network)	A network covering a small area, like a home, office, or school, where computers and devices share resources.	p. 142
Laptop Computer	A portable personal computer with a built-in screen and keyboard, capable of most tasks a desktop can do.	p. 34–35
Latency	The time it takes for a signal to travel from the source to the destination across a network.	p. 148
Logic Circuit	A collection of transistors arranged to perform a specific operation such as add, multiply, or compare.	p. 78
Machine Language	The language of computers, made up of binary signals (0s and 1s) that represent instructions and data.	p. 85
Magnetometer	A sensor in smartphones that detects the Earth's magnetic field to help determine orientation.	p. 58
Main Memory	The larger but slower memory (RAM) where programs and data are kept for active use.	p. 109
Messaging	A quick way of sending and receiving short pieces of information, often between friends or classmates.	p. 18
Modem	A device that converts digital computer signals into analog signals (and back) so they can travel over telephone lines.	p. 146
Multiplier	A logic circuit that multiplies two input signals and outputs the product.	p. 78
Multiplexer	A logic circuit that selects one signal from several possible inputs and forwards it.	p. 78
Network	A system of interconnected computers and devices that can exchange information.	p. 140
NIC (Network Interface Card)	A hardware card that allows a computer to connect to a network, available in Ethernet, fiber optic, or WiFi versions.	p. 146
Opcode (Operation Code)	The part of an instruction that tells the control unit which operation to perform.	p. 84
Operating System	A group of programs that manage how a computer works and allow other apps to run properly.	p. 20
Output Device	A device that takes internal computer signals and converts them into something external, such as sound, light, or movement.	p. 59
PAN (Personal Area Network)	A small network centered around one person's computer and devices, often using Bluetooth.	p. 140–141
Parallel Processing	When multiple tasks are truly processed at the same time, often by multiple cores or servers.	p. 175
Parallelism	A strategy that speeds up performance by performing multiple tasks at the same time, often using multi-core CPUs or GPUs.	p. 107
Parent Data	Information about students' parents or guardians, such as names and contact details.	p. 198

Performance	A measure of how fast and efficiently a computer completes tasks, influenced by factors like clock speed, cores, and memory design.	p. 106
Peripheral Device	An input or output device connected to a computer to extend its functions, like a printer, headset, or smartboard.	p. 124
Pointer	A small graphical symbol (like an arrow or hand) controlled by a mouse or touchpad to select and move items on the screen.	p. 63
Prediction	A design strategy where the processor guesses what instructions or data will be needed next and fetches them early.	p. 110–111
Prefetching	A prediction method where the processor not only fetches the needed data but also nearby data, assuming it will be used soon.	p. 111
Private Information	Information that is restricted and requires permission or a password to access.	p. 17
Processing (in IS)	The stage where raw data is organized, calculated, or analyzed to turn it into meaningful information.	p. 193
Processor (CPU)	The main component of a computer that changes input into output by executing instructions.	p. 76
Proximity	A design principle where components are placed physically closer together, like cache on the same chip as the processor, to speed up communication.	p. 109
Public Information	Information that is open to everyone and does not require a password.	p. 17
Redundancy	Keeping extra copies of data so it isn't lost if transmission errors occur.	p. 151
Register	A very small and fast memory location inside the processor, used to hold data temporarily.	p. 88
Register File	A group of registers managed together within the processor to store data and addresses close to the ALU.	p. 88, 109
Repeater	A device that boosts weakened signals so they can travel longer distances without losing strength.	p. 151
Response Time	The total time it takes for a user to send a request and receive a response back from a server.	p. 176
Robot	A machine with sensors and actuators that can perform tasks either autonomously or with some level of human control.	p. 128
Router	A device that directs signals between networks, acting as both a gateway to the internet and a traffic controller.	p. 143
Satellite Imagery	Images of Earth taken from satellites, used in GIS to provide wide-area geographic information.	p. 199
School Information System (SIS)	An information system that manages school-related data such as schedules, grades, attendance, and communication with students, parents, and staff.	p. 193
Semi-Autonomous Robot	A robot that can make some decisions on its own but still relies partly on human guidance.	p. 128

Server	A powerful computer in a data center that stores data or runs applications to provide services for other computers and users.	p. 38, 173
Shared Cache	A cache memory used by multiple cores in a processor, storing data that any of them may need.	p. 110
Shielding	A protective layer around wires or circuits to block outside interference.	p. 151
Shifter	A logic circuit that moves the bits of a signal left or right by a certain number of places.	p. 80
Signal	A message carried in physical form, such as electrical energy, that allows parts of a system to communicate and coordinate.	p. 55
Smart Device / Smart Thing	A device with sensors and connectivity that links to a network (like smart watches, fridges, or cameras) to automate or improve tasks.	p. 179
Smart Farming	The use of IoT devices such as soil sensors, drones, and livestock trackers to optimize farming practices.	p. 180
Smart Home	A home equipped with IoT devices (like smart lights, thermostats, or security cameras) connected to a hub that automates daily tasks.	p. 179
Smart Machine	A machine with an embedded computer that adds new features and makes it more functional and user-friendly.	p. 126
Smartphone	A general-purpose personal computer in the form of a phone, capable of running many applications beyond calls and texts.	p. 126
Smart Waste Management	A city system using IoT sensors in bins and trucks to optimize garbage collection and recycling.	p. 181
Software (in IS)	Programs and applications, including databases, that process and manage data within an information system.	p. 197
Source Address	The location where input data for an instruction is stored before processing.	p. 82
Specific-Purpose Computer	A computer designed to perform one set of tasks very well, such as a supercomputer, server, or embedded computer.	p. 36
Specific-Purpose Machine	A machine designed to perform only one type of task, like a cash register at a store.	p. 127
Storage (in IS)	The function of keeping collected data in databases, usually on servers or in the cloud, so it can be accessed later.	p. 192
Storage	The largest and slowest form of memory, used for long-term keeping of data and programs (e.g., hard drives or SSDs).	p. 109
Stored Program	The concept that a computer keeps instructions inside it, allowing it to perform different tasks by running different programs.	p. 19–20
Street View Data	360° street-level images collected by special vehicles for use in GIS platforms like Google Maps.	p. 199
Student Data	Information about students, including names, IDs, grades, test scores, and attendance.	p. 198

Supercomputer	A massive and very powerful computer built to solve one very complex problem at a time, such as weather forecasting or scientific research.	p. 36–37
Synchronization	Keeping all computer components working together in step using timing signals from the clock.	p. 104
System	A set of interconnected parts working together to serve a particular purpose.	p. 54
System Administrator	A professional who oversees and maintains an information system, manages user accounts, installs updates, and ensures smooth operation.	p. 195
Tablet	A portable personal computer that uses a touchscreen instead of a physical keyboard, making it lighter but usually less powerful than a laptop.	p. 34–35
Teacher Data	Information about teachers, such as their names, subjects taught, schedules, and contact details.	p. 198
Technology	Tools, machines, or systems created to help us perform tasks, from pencils to computers.	p. 10
Throughput	A measure of server performance, defined as the number of requests a server can complete in one second.	p. 174–175
Usability	How easy it is for people to use a computer or interface to complete a task.	p. 64
User Interface (UI)	The software part of the human-computer interface that lets users give instructions to a computer and see results.	p. 61
Voice User Interface (VUI)	A user interface where people interact with computers by speaking, using natural language commands.	p. 63
WAN (Wide Area Network)	A larger network covering a city, country, or even the whole globe. The internet is the biggest WAN.	p. 143–144
Web Address	A special type of address used to locate a file or webpage on the World Wide Web.	p. 17
WiFi NIC	A network card that connects a computer to wireless networks using radio signals.	p. 146
WIMP Interface	A type of GUI that uses Windows, Icons, Menus, and a Pointer to interact with the computer.	p. 63
Word (in computing)	The largest group of bits a processor can handle at once; in a 32-bit processor a word is 32 bits, in a 64-bit processor it is 64 bits.	p. 77
World Wide Web (Web)	The collection of files, websites, and content (text, images, videos) that can be accessed through the Internet.	p. 17

INDEX

Accelerometer – Sensor in smartphones that detects linear motion in three dimensions: up/down, left/right, forward/backward (p.58).

Actuator – An output device capable of performing an action, such as a motor, valve, or printer head. Converts internal digital signals into external movement (p.59, 72).

Action potentials – Electrical signals used internally by humans to process information (p.122–123).

Adapters – Hardware devices that convert internal digital signals of a computer into external transmission signals (p.145).

- Wired adapters – Land line modems, Ethernet NICs, Fiber optic NICs (p.146).
- Wireless adapters – Cellular modems, WiFi NICs, Bluetooth adapters (p.146).

Adder – A logic circuit that performs binary addition (p.78–79).

Address – Used to locate a file on a computer (p.19). In processors, source and destination addresses specify where data is fetched from or written to (p.82–83, 89–90). See also Web address.

Acknowledgement – Signal sent by a receiver to confirm successful receipt of a transmission (p.152).

ALU (Arithmetic Logic Unit) – The part of the processor responsible for performing arithmetic and logical operations using logic circuits (p.81–82).

Applications (Apps) – Programs stored on a computer that perform specific tasks (p.20–21).

Architecture – The basic structure of a computer, consisting of processor, memory, input/output devices, a bus, and a clock (p.102).

ASCII values – Numerical representation of characters used in checksum calculations (p.166–167).

Attenuation – Weakening of a signal as it travels over long distances (p.150).

Autonomy (robots) – The degree to which a robot can act independently; can be autonomous, semi-

autonomous, or non-autonomous (p.128).

Bandwidth – Amount of data transmitted per second, measured in bits per second (bps); critical for throughput and server performance (p.175–176).

Barcode scanner – Input device used to scan product barcodes, providing information such as name and price (p.127).

Biotechnology – Technology that uses biological processes, cells, and organisms to make products (p.24).

Binary signals – Digital electronic signals that can only have two values: high voltage (1) or low voltage (0) (p.77).

Bluetooth – Wireless short-range communication technology; Class 2 antennas reach 10 m, Class 1 antennas up to 100 m (p.141, 146).

Bus system – A shared communication pathway that connects all components of a computer. Includes the data bus, address bus, and control bus (p.103).

Cache – A memory component placed close to the processor to speed up access to data and instructions (p.109–110).

- Instruction cache (I-cache) – Stores instructions for quick access (p.110).
- Data cache (D-cache) – Stores data items for quick access (p.110).
- Shared cache – Serves multiple cores on a processor (p.110).

Caesar cipher – Early encryption method shifting letters by a fixed number of positions (p.154–155, 168–169).

Capability – The complexity of tasks a computer can perform, determined largely by computational power (p.34).

Cash register – Machine with an embedded computer designed for point-of-sale tasks; includes barcode scanner, keyboard, screen, drawer, and printer (p.126–127).

Checksum – Error detection method where values of transmitted data are summed and compared (p.152, 166–167).

CLI (Command-Line Interface) – A text-based user interface where users type commands to interact with the computer. Common in programming and system administration (p.61).

Clock – Synchronizes activities in a computer by generating regular timing signals. Measured in Hertz (Hz); higher clock frequencies increase performance (p.104–106).

Cloud – System of servers accessed over the internet that provide storage and processing services (p.170–172).

Communication – Exchange of information between two or more people (p.18).

- Email – Formal or longer communication method (p.18, 21).
- Messaging – Quick, short exchanges of information (p.18, 21).

Comparator – A logic circuit that compares two inputs and outputs greater than, less than, or equal (p.80).

Computers – Technology designed to process information according to a stored program (p.10, 15).

- Creating information – Producing new content through writing, recording, or calculation (p.15–16).
- Sharing information – Exchanging content via Internet, messaging, or email (p.16–18).
- Storing information – Keeping files permanently for retrieval (p.19).
- Types of computers – Classified as general-purpose or specific-purpose (p.32).

Control Unit (CU) – Directs the flow of signals through the processor, decodes instructions, generates control signals, and coordinates components (p.84–86).

Cybersecurity – Field of protecting computer systems, networks, and data from unauthorized access and attacks (p.155).

Data – Raw values that can be processed to create new information (example: student grades) (p.15–16).

Data centers – Facilities with large numbers of servers

and supporting systems (switches, cooling, power, security) that handle heavy workloads and provide cloud services (p.38–39, 173–177).

- Dependability – Ability to operate without fault or failure (p.177).
- Performance measures – Throughput, bandwidth, response time (p.174–176).

Data transmission errors – Errors caused by attenuation and interference (p.150).

Database servers – Servers that manage and store data in databases (p.40).

Dependability (systems) – The ability of a system to operate without fault or failure; in data centers can be affected by overheating, outages, breaches, or disasters (p.177).

Desktop computer – Large, stationary personal computer with high capability but low portability (p.34–35).

Digital electronic signals – Electrical signals used by computers for internal information processing. Binary in nature (1 = high voltage, 0 = low voltage) (p.55, 77).

Directory – A logical structure used by the filing system to organize files. Also called a folder; can contain files and subdirectories. Represents a hierarchical structure (p.65, 73–75).

Embedded computers – Computers built into other devices, such as cars, pacemakers, or appliances, designed for energy efficiency, small size, and high reliability (p.41–42).

- Examples – Pacemaker, car engine control, home appliances, airplanes (p.42, 49).
- Other names – Control modules, controllers, microcontrollers (p.49).

Email servers – Handle email accounts and deliver messages (p.40).

Encryption – Securing data by converting plaintext into unreadable ciphertext using an algorithm and a key (p.153–155).

Ethics (robots) – Concerns about whether robots will act responsibly, ethically, and in humans' best interests (p.130).

Execution time – The total time a CPU takes to

complete a program. A measure of performance (p.112, 117).

External drives – Storage devices such as HDDs or SSDs that provide large capacity but risk damage or loss (p.171–172).

Fiber optic cables – Backbone of the internet, transmitting light signals across continents (p.144).

Fields (Instruction) – Sections of a machine language instruction, typically including opcode, data, and addresses (p.87).

File – A unit of storable information (p.19).

- Filename – Label assigned to identify a file (p.19).
- File servers – Servers that store and share files (p.40).

Game servers – Host online games and manage player accounts (p.40).

General-purpose computers – Flexible computers that run a wide range of applications; include personal computers such as desktops, laptops, tablets, and smartphones (p.32–33).

Geographic Information System (GIS) – Information system for collecting, storing, processing, and sharing geographic data, often visualized as maps (p.194–196).

Graphical User Interface (GUI) – Uses windows, icons, and menus to make computers more intuitive and accessible (p.62).

- WIMP Interface – GUI controlled by mouse or touchpad with windows, icons, menus, and pointer (p.63).
- Touchscreen Interface – GUI where users interact directly with the screen through gestures like tapping or swiping (p.63).

GPU (Graphics Processing Unit) – Specialized processor that executes graphics-related instructions. Works in parallel with the CPU (p.107).

Gyroscope – Smartphone sensor that detects rotational movement: clockwise/counterclockwise, rolling, pitching (p.58).

Hardware – The physical components of a computer, including input and output devices, that handle the physical (signal) layer of the interface (p.60).

Hop count – The number of routers a signal passes

through on its path to a destination (p.147–149).

Human-Computer Interaction (HCI) – A multidisciplinary field studying how humans interact with computers and other technologies (p.122).

Human-Computer Interface – The boundary where humans and computers interact via input/output devices (p.122–123).

Human-Machine Interface – The boundary where humans interact with machines, often through embedded computers (p.126).

I-cache (Instruction cache) – Cache memory dedicated to storing instructions for faster execution (p.110).

Information – Meaningful content about ourselves or the world, processed and used by humans and computers (p.12–13).

- Creating (p.15).
- Sharing (p.16–18).
- Storing (p.19).

Information systems – Systems that collect, store, process, and share data to support decision-making (p.192).

- Functions – Collection, storage, processing, sharing (p.192).
- Components – Hardware, software, data, people (p.197–199).
- Examples – School Information Systems (SIS) (p.193–195); Geographic Information Systems (GIS) (p.194–196).

Input devices – Devices that convert external signals (light, sound, pressure) into internal signals; e.g., keyboard, mouse, microphone, camera, touchscreen, sensors (p.57–58).

Input/Output (I/O) loop – Continuous cycle of input and output exchanged between humans and computers until a task is completed or interrupted (p.123).

Instruction cycle – The repeated process of fetch, decode, and execute, performed continuously by the processor (p.91).

Instruction set – The complete set of instructions a processor is designed to perform (p.83, 92).

Interface – The boundary at which a system interacts with the external world by converting signals (p.56).

- User Interface (UI) – Software layer translating user actions into computer instructions (p.61).

Interference – Disruption of signal transmission caused by storms, buildings, electromagnetic radiation, or cosmic rays (p.150).

Integrity of transmission – Assurance that messages are delivered error-free and unchanged (p.149–150).

Internet – A global telecommunications network that connects computers and devices (p.16–17).

Internet of Things (IoT) – System of interconnected devices (“things”) that send or receive data over networks, often autonomously (p.178–181).

- Smart homes – Devices like lights, thermostats, and appliances connected via hubs (p.179).
- Smart farms – IoT used in agriculture with soil sensors, drones, and livestock monitors (p.180).
- Smart cities – IoT for waste management, traffic control, and public safety (p.181).

Keyboard – Input device used to enter text or numbers into a system (p.127).

Laptops – Portable personal computers with capabilities close to desktops, powered by rechargeable batteries (p.34–35).

Latency – The time it takes for a signal to travel from source to destination (p.148–149).

Local Area Network (LAN) – A network covering a limited area like a school or office, usually wired (p.142–143).

Logical layer – The information carried by signals; handled by software in the human-computer interface (p.60).

Logic circuits – Collections of electronic components (mainly transistors) arranged to perform operations such as addition, multiplication, shifting, comparison, and selection (p.78–80).

Magnetometer – Smartphone sensor that detects magnetic fields and orientation relative to Earth’s magnetic field (p.58).

Maps (GIS data) – Digital or paper maps used as part of geographic data (p.199).

Memory cards – Expandable portable storage for devices like smartphones; limited capacity and risk of damage (p.171).

Memory system – Hierarchical system that supplies data and instructions to the processor. Components include storage, main memory, cache, and register file (p.109).

Mobility (robots) – A robot’s ability to move; stationary (factory arms) vs. mobile (vacuums, humanoids) (p.128).

Modems – Devices that convert between digital and analog signals for transmission (p.146).

Multiplier – A logic circuit that multiplies two inputs (p.78–79).

Nanotechnology – Manipulation of matter at the nanoscale (1–100 nm) to make products (p.24).

Networks – Systems of interconnected computers and devices capable of exchanging information (p.140).

- PAN (Personal Area Network) – Smallest network, centered on an individual’s devices (p.140–141).
- LAN (Local Area Network) – Covers homes, offices, or campuses (p.142–143).
- WAN (Wide Area Network) – Largest networks, including the internet (p.143–144).

NIC (Network Interface Card) – Hardware adapter that connects computers to networks (p.146).

Opcode (Operation code) – The part of an instruction that specifies the operation to perform. Decoded by the control unit (p.84–85).

Operating System – Group of stored programs that allow a computer to function properly (p.20).

Output devices – Convert internal signals into external signals (e.g., sound from speakers, light from screens) or actions (via actuators). Includes printers, screens, drawers, projectors, and more (p.59, 125, 127).

Pacemaker – Example of an embedded computer in medicine; regulates heartbeats (p.42).

Parallel processing – Running multiple tasks simultaneously to enhance performance, used in CPUs and clustered servers (p.175–176).

Parallelism – A performance optimization strategy where tasks are performed simultaneously, often

through multi-core CPUs and GPUs (p.107).

Performance measures – Metrics such as clock frequency, instruction count per second, and program execution time (p.106, 112).

Peripherals – External input, output, or I/O devices connected to computers to extend functionality (p.124–125). Examples: printer, projector, smartboard, webcam, scanner, headset.

Personal computers (PCs) – General-purpose computers for individual use, including desktops, laptops, tablets, and smartphones (p.34–35).

- Distinguished by portability (size, weight, power source) and capability (computational power) (p.34).
- Designed to balance flexibility, usability, capability, and affordability (p.35).

Physical layer – The actual signals transmitted and received, handled by hardware (p.60).

Ping – Networking tool to measure round-trip time between source and destination (p.148–149).

Portability – The ease with which a computer can be carried, depending on size, weight, and power supply (p.34).

Prediction – Optimization strategy that anticipates the next instruction or data likely to be needed, using separate caches or prefetching (p.110–111).

Prefetching – A prediction strategy where the processor fetches adjacent data items in advance, improving performance (p.111).

Printers – Output devices producing receipts or documents; use actuators like motors and printheads (p.127).

Processing – Structured change from one form to another, carried out by a series of repeatable steps (p.11–13).

- Information processing – Comparison with memory to identify and interpret (p.13–15).

Processing (cloud-based) – Service where computational power is provided via the cloud rather than on local devices (p.172).

Processor (CPU) – The central unit that changes input into output. Composed of ALU, CU, and register file

(p.76, 92).

Program – A set of instructions to perform a specific task (p.20–21).

- Stored program – Concept that made modern computers possible (p.19–21).

Projector – Peripheral output device that displays visuals for large audiences (p.125).

Public vs. Private Information –

- Public – Accessible to everyone (p.17, 21).
- Private – Restricted by password access (p.17, 21).

Purpose (robots) – The specific tasks robots are designed to perform, from repetitive labor to hazardous work (p.128–129).

Redundancy – Strategy of keeping multiple copies of information to protect against loss (p.151).

Registers / Register file – Small, fast memory components located within the processor, managed by the CU, used to hold data and addresses during execution (p.88–90, 92).

Repeaters – Devices that amplify and retransmit weakened signals to prevent loss (p.151).

Research – Task example for general-purpose computers, such as using a web browser (p.33).

Response time – The time it takes for a server to respond to a user's request, as seen from the user's perspective (p.176).

Robots – Machines capable of performing tasks autonomously or with human intervention, using sensors (input) and actuators (output) (p.128–130).

- Benefits – Perform repetitive or dangerous tasks (p.129).
- Concerns – Job displacement, ethical behavior (p.130).

Routers – Devices that connect LANs to WANs, serving as gateways and traffic controllers; determine best routes using hop count and latency (p.143–149).

School Information Systems (SIS) – Information systems for managing school operations; provide schedules, attendance, grades, communication, and reports (p.193–195).

Screen (cash register) – Output device displaying transaction details (p.127).

Sensors – Input devices that allow robots, IoT, and GIS to collect environmental or situational data (p.128–129, 179–181, 199).

Servers – Computers designed to handle many simpler tasks concurrently, often for sharing or processing information (p.38–40).

- Database servers – Manage and store data (p.40).
- Email servers – Manage accounts and deliver mail (p.40).
- File servers – Store and share files (p.40).
- Game servers – Host games and manage players (p.40).
- Web servers – Host websites and deliver web pages (p.40).
- Characteristics – Optimized for performance, reliability, and security (p.41).

Shielding – Protection for copper wires and satellites against interference (p.151).

Shifter – A logic circuit that shifts bits of input left or right by a specified number of places (p.79–80).

Signals – Messages carried in physical form, such as electrical potential energy in computers (digital electronic signals) or humans (action potentials) (p.55).

Smart machines – Machines with embedded computers that enhance interfaces and add functionality (p.126).

Smartboard – I/O device that displays content and allows direct manipulation (p.125).

Smartphone – A portable general-purpose computer evolved from cellphones. Combines communication (calls, messaging) with computing and internet functions. Considered the smallest, most portable type of personal computer, capable of many tasks (p.34–35, 126).

Software – Programs and databases that determine computer behavior and manage information. In general, software handles the logical layer of the interface (p.60). In information systems, it includes applications and databases that collect, store, process, and share data; these may be single applications or multiple integrated apps (p.197–198).

Specific-purpose computers – Designed for a defined

set of functions, e.g., supercomputers, servers, and embedded computers (p.36–43).

Storage – Permanent repository for information (p.19).

Storage options – Memory cards, external drives (HDD, SSD), and cloud-based storage; each with advantages and limitations (p.171–172).

Street View (GIS data) – Images captured by vehicles with cameras, providing street-level perspectives (p.196, 199).

Supercomputers – Large, powerful computers designed to solve one complex problem at a time (p.36).

- Example – IBM Blue Gene, used for weather forecasting (p.37).
- Uses – Scientific research, weather forecasting, oil and gas exploration, molecular modeling (p.36–37).

Switches – Devices connecting multiple computers and peripherals within a LAN (p.143).

Synchronization – Strategy that ensures components operate in unison using a clock signal, preventing errors and delays (p.104).

Systems – Sets of interconnected components working together for a purpose. Humans and computers are both information-processing systems (p.54).

Tablets – Portable personal computers larger than smartphones but smaller than laptops, using touchscreen technology (p.34–35).

Technology – Tools, machines, or applications of knowledge designed for a purpose (p.10–11).

- Transportation technologies – e.g., car, airplane, ship (p.23–24).
- Other domains – Nanotechnology, Biotechnology (p.24).

Three Laws of Robotics – Ethical rules proposed by Isaac Asimov: robots must not harm humans, must obey humans, and must protect themselves unless it conflicts with the first two laws (p.130).

Throughput – Measure of server performance, defined as the number of requests processed per second (p.174–175).

Transistors – Electronic components that form the

building blocks of logic circuits; their arrangement determines operations (p.78).

Trust in networks – The expectation that transmissions are accurate, secure, and private (p.149–150).

Usability – The ease with which a device or interface can be used to complete a task. Higher in GUIs and touchscreens, lower in CLIs (p.64, 126).

User Interface (UI) – The software component of the human-computer interface, translating input into instructions and output into results (p.61).

- CLI, GUI, VUI – Major types of user interfaces (p.61–63).

Voice User Interface (VUI) – Interface that allows interaction using voice commands, e.g., Siri, Alexa, Google Assistant (p.63).

WAN (Wide Area Network) – Large networks covering cities, countries, or the globe; the internet is the largest example (p.143–144).

Web (World Wide Web) – Internet-based information system of files and pages (p.17).

- Web address – Location identifier for content on the Web (p.17, 19).
- Web servers – Host websites and deliver web pages (p.40).

Webcam / Camera – Input devices that capture images or video, extending the computer interface (p.125).

WiFi – Wireless networking technology for LAN and internet connectivity; typical range 45–150 m (p.146).

WIMP Interface – GUI type using windows, icons, menus, and pointer controlled by mouse/touchpad (p.63).

Word (processor architecture) – The maximum number of bits transmitted simultaneously (e.g., 32-bit word, 64-bit word) (p.77).

